Recurrent Adaptive Graph Reasoning Network With Region and Boundary Interaction for Salient Object Detection in Optical Remote Sensing Images

Jie Zhao, Yun Jia, Lin Ma, *Senior Member, IEEE*, and Lidan Yu

***Abstract*— In the realm of optical remote sensing imagery, tackling the intricate task of detecting salient objects poses challenges that expose limitations in prevailing CNN- and transformer-based methodologies, particularly in dealing with intricate object topologies. To address this, we present a novel solution, the recurrent adaptive graph reasoning network (RAGRNet), tailored to capturing patterns of salient objects within complex and irregular remote sensing images. For the first time, our network orchestrates simultaneous modeling of both object regions and boundaries, ushering in a comprehensive comprehension of their intrinsic correlations. This pioneering approach involves the deployment of a multigranularity boundary-focused module (MGBF), which effectively crafts enriched boundary features. Going further, the recurrent adaptive graph reasoning (RAGR) module scrutinizes the semantic interplay between region and boundary representations, thereby fortifying the learning process of intricately structured object features. Concluding the process, a specialized decoder, termed the heterogeneous semantic cooperative interconnection (HSCI), systematically fuses region and boundary information in a progressively unified manner. This intricate interplay harnesses their synergistic complementarity to bolster the Precision and sophistication of modeling salient objects. In rigorous evaluations across diverse datasets, our proposed network outperforms 38 state-of-the-art (SOTA) methodologies, a testament to its remarkable efficacy. Code and results will be available at https://github.com/JieZzzoo/RAGRNet.**

***Index Terms*— Adaptive interaction, graph convolutional network (GCN), multigranularity, optical remote sensing images, salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) constitutes an indispensable and integral facet of computer vision tasks. Its principal objective is to emulate the human visual attention mechanism, enabling accurate localization and segmentation of salient entities within images. In the current landscape, SOD has garnered widespread applicability across a multitude of domains. These domains span image quality assessment, object detection, semantic segmentation, visual tracking, and beyond. Consequently, it has cemented its position as a foundational element that underlies numerous visual analytical pursuits.

In the initial stages, SOD predominantly relied on manual feature extraction methods, such as emphasizing objects based on attributes like color, texture, and shape. However, this approach encountered limitations in capturing higher level semantic information, leading to suboptimal saliency results. Benefitting from the rapid advancements of fully convolutional networks (FCNs) [1], significant strides have been made in the field of salient detection. Notably, researchers such as [2] introduced a boundary-guided perceptual approach, achieving precise segmentation outcomes for intricate structures and high-fidelity boundaries. Similarly, techniques like [3] harnessed multiscale information capture to enhance overall region-based saliency, while [4] proposed a strategy to mine vital information from challenging samples, mitigating noise interference in images within an unsupervised context. Simultaneously, the work by Fan et al. [5] pioneered camouflage target detection, complementing saliency detection by highlighting concealed objects in specific environmental contexts. Beyond individual image inputs, success has also extended into the realm of multimodal contexts. This phenomenon is evident in scenarios involving RGB and depth/thermal/infrared images (RGB-D/T/I) [6], [7], [8], as well as video saliency detection [9]. Following this, SOD in optical remote sensing images (ORSI-SOD), a highly intricate and challenging domain analogous to the aforementioned SOD in nature sensing images (NSI-SOD), has garnered significant scholarly interest. ORSIs, captured from aerial perspectives by satellite equipment, typically manifest three distinct characteristics: first, they involve multiple objects of varied sizes, with smaller objects often prevailing in the scene; second, these objects possess intricate topological structures and elongated features; and third, objects are situated within complex environments, presenting challenges in distinguishing them from their backgrounds. These inherent challenges impose stringent demands on saliency models. It is noteworthy that even the top-performing methods in natural scene SOD, such as F3Net [10] and PA-KRN [11], face limitations, as depicted in Fig. 1. Their significant disparities from authentic saliency object images, particularly in the realm of small objects and complex topological structures, underscore the impracticality of directly transferring NSI-SOD

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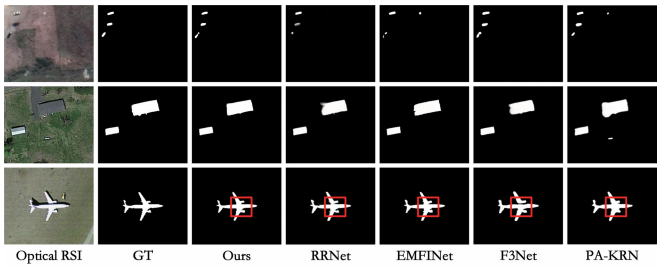


Fig. 1. Visual assessment was conducted on saliency maps produced by our proposed RAGRNet, along with those generated by RRNet [20], EMFINet [21], F3Net [10], and PA-KRN [11]. Instances, where distinctions were challenging to discern, were demarcated using red rectangular annotations. For a detailed examination, we recommend a zoomed-in view of the saliency maps.

methods to the context of ORSI-SOD. Nevertheless, these methodologies have played a pivotal role in propelling the field of ORSI-SOD forward.

Given the distinct characteristics of ORSIs, many scholars have drawn upon the insights from NSI-SOD and devised methods to propel the advancements in this field. For instance, LVNet [12] and MCCNet [13] construct multiscale information architectures to facilitate effective context propagation. PDFNet [14], SARNet [15], and CorrNet [16] thoroughly exploit the multilayered complementarity extracted from CNNs [18]. DAFNet [19] extends the attention mechanism from shallow to deep layers, harnessing context information for modeling long-range relationships. Similarly, RRNet [20] builds on a multiscale attention framework and employs relational evolution to capture irregular information representations. Furthermore, approaches such as EGNet [2] enhance the performance of SOD by integrating boundary information. However, while these methodologies drive advancements, they also exhibit certain limitations. RRNet [20], for example, is capable of inferring internal relationships within salient objects, yet its reasoning is confined to local regions, disregarding the boundary information's role in determining global relationships. EMFINet [21], despite its multiscale information extraction, overlooks the inherent relationships of objects, leading to issues in overall representation, as depicted in Fig. 1. In light of these observations, we delve into a novel approach that simultaneously considers both region and boundary information for relational reasoning. This approach fosters structured relationships among various salient objects, both interobject and intraobject, thereby compensating for the information gaps and enhancing the overall comprehension.

To address the existing challenges in ORSI-SOD, we introduce a novel recurrent adaptive graph reasoning network (RAGRNet). This network employs interactions between region and boundary features to fuse and refine heterogeneous information, thereby enhancing the detection of salient objects within intricate remote sensing scenarios. Concretely, we establish an encoder–decoder framework with the encoder incorporating prevalent feature extraction backbone networks. Building upon this foundation, we devise three distinct modules. First, a multigranularity boundary-focused module (MGBF) is designed to amplify and enrich boundary feature representations across varying scales. Second, a recurrent adaptive graph reasoning (RAGR) module is constructed to facilitate collaborative learning between target and contextual scenes, enhancing the intrinsic correlations within boundary and region representations. Finally, we introduce a decoder named heterogeneous semantic cooperative interconnection (HSCI), which, via joint abstraction representations from multiple levels of information, promotes refined interaction within the decoder, ultimately delineating foreground targets.

The primary contributions can be summarized as follows.

1) We propose an end-to-end network, i.e., RAGRNet, to tackle the challenges of ORSI-SOD. The RAGRNet capitalizes on the progressive fusion of diverse sources of heterogeneous feature information, and cyclic updates of node features coupled with graph reasoning augment interobject relationships.

2) Traditional boundary refinement methods are incorporated into the RAGR, enhancing the latent relationships between refined boundary information and region information nodes, thereby enriching structured representations of target features. Notably, the refined boundary information emanates from the MGBF.

3) The HSCI is established, harnessing the synergy between boundary and region features. While achieving information distillation and utilization, it effectively decodes salient foreground targets.

The structure of this article is organized as follows. We review relevant research pertaining to SOD and graph reasoning networks in Section II. Section III provides a comprehensive exposition of the RAGRNet proposed in this study. In Section IV, we conducted a comprehensive analysis of module functionality, delving into typical cases of faults and exploring the adaptability of large-area detection. The concluding section summarizes the entire body of work.

II. Related Works

In this section, we will provide a retrospective analysis of NSI-SOD, ORSI-SOD, and graph reasoning networks, respectively, along with their applications.

A. SOD for NSIs

The advancement of NSI-SOD has rapidly gained prominence in the field of computer vision over the past decade. This progression has undergone a transformation from manually engineered features to the automatic extraction of features through CNN-based approaches. In its early stages, various methodologies explored the advantages of factors such as region contrast [22], [23], contextual cues [24], object priors [25], [26], and background knowledge [27], [28] in enhancing saliency detection. For instance, Achanta et al. [22] introduced a frequency-tuned method for estimating contrast around central regions while preserving spatial information. Optimization frameworks were employed in [25] to combine multiple low-level cues to depict the spatial layout of image regions relative to boundaries, and in [28] to achieve foreground-background separation through background modeling, leveraging unsupervised learning for more comprehensive

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data representations. Another approach, proposed in [24], utilized a Boolean-based BMS model to analyze random threshold binary maps of images, calculating saliency maps through topological structure analysis. While these traditional techniques might not have achieved remarkable performance, they have offered thought-provoking solutions for NSI-SOD.

Leveraging the potent feature extraction capabilities of CNN in image analysis, researchers have seamlessly integrated them into NSI-SOD, yielding remarkable strides in performance enhancement [2], [29], [30], [31], [32], [33], [34], [35], [36]. In the approach by Hou et al. [29], a top-down strategy was employed, reinforcing shallow-level object regions through succinct connections. To ensure the inclusivity of larger objects, [30] employed a multilevel pooling technique to augment the receptive field dimensions. An instantly deployable dense attention module was introduced in [32], adeptly preserving the complete salient object and its boundaries. Through boundary-aware mechanisms, [2], [31] meticulously refined texture features, assisting saliency maps in capturing holistic information. Strategies for contextual interaction were harnessed in [33] and [34] to amplify the network's grasp of the overall scene, steering lower levels to assimilate pertinent features. Parallelly, [35] enhanced U-shaped architectures through stacking, optimizing the response of diverse feature channels to foreground and background elements. In contrast to CNN-centric approaches, [37] embraced the T2T-ViT [38] framework, crafting saliency detection networks suitable for both RGB and RGB-D scenarios, offering an efficient global information modeling. Drawing upon transformer encoders, [39] captured extended contextual dependencies, preserving global integrity with accuracy. Employing a multirelational edge convolution operation, [36] methodically refined graph structures layer by layer, aggregating interconnected node information and rivaling CNN performance. The aforementioned CNN-based NSI-SOD models have not only achieved state-of-the-art (SOTA) performance but have also provided valuable impetus for ORSI-SOD.

B. SOD for ORSIs

Emerging as a burgeoning field within SOD, ORSI-SOD has evolved into an exceptionally challenging task within remote sensing analysis. Similar to NSI-SOD, early approaches also centered on crafting hand-engineered features to quantify ORSIs, including aspects like color, contrast, and sparse representations. Zhang et al. [40] constructed histogram mapping color information within the color dimension, effectively establishing saliency maps through the fusion of saliency scores. Drawing from the concept of dual dictionaries, [41] which, respectively, encapsulate local and global representations, sparse representations were harnessed to fuse global and background cues, culminating in the production of saliency maps. Zhang et al. [42] introduced an innovative multicue adaptive fusion strategy, accommodating inherent relationships by incorporating color, texture, and global contrast features within the context of ORSI. Furthermore, the integration of machine learning techniques has been explored. As an example, in a study by Zhang and Yang [43], a frequency-domain analysis approach utilizing quaternion Fourier transform was applied, building upon HIS transformation to accelerate saliency map generation. In another method presented by Liu et al. [44], an unsupervised saliency model incorporating a color Markov chain (US-CMC) was introduced, specifically designed for the detection of oil tanks, enhancing the detectability of real targets in subjective detection scenarios. Moreover, Zhang and Zhang [45] proposed the fusion of wavelet transform with intensity-hue-saturation, catering to distinct saliency requisites across various regions. Nonetheless, the adoption of these methods remains limited due to the inherent limitations of generating fixed features and achieving only finite levels of saliency Precision.

Significant advancements have been achieved in the field of ORSI-SOD, thanks to the evolution of deep learning and the accessibility of extensive datasets like ORSSD [12], EORSSD [19], ORSI-4199 [46], and RSSOD [47]. Early on, CNN-based approaches started tackling challenges associated with multiscale remote sensing objects, irregular topological structures, and intricate boundaries. Broadly, these methods can be categorized into two groups: those that enhance semantic information and those that amalgamate boundary-assisted semantic information. To illustrate, LVNet [12] employs a dual-stream pyramid structure to iteratively refine semantics, DAFNet [19] employs attention mechanisms during encoder-based feature extraction to eliminate noise and convey nuanced information to the deeper network, bolstering semantic expression. Meanwhile, ACCoNet [48] introduces a context-interaction mechanism to distill primary encoded information, thereby facilitating improved decoding. In contrast, the second category involves networks that integrate diverse content sources, with a distinct emphasis on boundary-assisted semantic information. For instance, ERP-Net [49] employs a recursive strategy to precisely position salient areas using boundary guidance. MJRBM [46] maximizes the representation of boundary and region information through a cross-promoting pattern, utilizing gating mechanisms. MCCNet [13] supplements boundaries, foreground, edges, and background through multifaceted interactions, effectively accentuating salient regions across various scales under the guidance of attention mechanisms. Furthermore, some networks draw inspiration from Transformers and integrate long-range self-attention mechanisms into ORSI-SOD, thereby enhancing the representation of global information. ASTT [50] introduces a detection network that is exclusively built upon transformers, delving into both global and local detailed information. In a manner akin to ASTT [50], GLGC-Net [51] also incorporates transformers but distinguishes itself by employing boundary-assisted and multiscale parallel semantic perception modules for context-aware saliency modeling. Taking a different route, [52] puts forward a dual-branch architecture that combines CNN and transformer, facilitating interactive guided aggregation of multilevel global and local features. In parallel with the aforementioned theoretical investigations, [53], [54] advanced theories and methodologies have been introduced for large-area SOD, particularly in the context of urban green space (UGS). Nevertheless, prevailing

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methodologies continue to encounter challenges in effectively capturing irregular topological structures.

C. Graph Reasoning Networks

Graph neural networks in the early forms can be traced back to references [55] and [56]. Micheli [57] introduced a preliminary version of a spatial-based graph convolutional network (GCN) by integrating composite nonrecurrent layers into the architecture. Subsequently, researchers extensively applied GNN to various domains such as knowledge graphs [58], social networks [59], protein interactions [60], and recommendation systems [61], [62]. Similarly, its presence can also be observed in image classification and semantic segmentation tasks. Jia et al. [63] proposed an intrasuperpixel and intersuperpixel structure, corresponding to local and global features of ground objects. Wan et al. [64] constructed a dynamically multiscale aggregation network, which dynamically updates the graph structure throughout the graph convolution process, resulting in refined and discriminative graph representations. Fabijanska et al. [65] took a fresh look at image segmentation, reframing it as a semi-supervised node classification task within the adjacency graph of superpixel regions to overcome the costly annotation process. Within the context of ORSI-SOD, some studies have also harnessed the benefits of graph-based reasoning capabilities. Cong et al. [20] introduced a relational reasoning network that pinpoints semantic relations in both the channel and spatial dimensions, concurrently encoding information for reuse across multiple scales through essential features. Similarly, UG2L [66] incorporated the advantages of graph-based reasoning into the end of the encoding process, refining position information generated by higher level semantics. However, these methods efficiently utilized only regional information, failing to fully exploit the complementary nature of regional and boundary representations, which consequently led to suboptimal expression of topological structures. In this article, we construct a more comprehensive graph structure, simultaneously integrating object region and boundary information into graph nodes. This approach captures the semantic relationships between regions and contours, facilitating the flow of information among salient regions and achieving a more precise representation of topologically significant objects.

III. Proposed Method

In this section, our proposed RAGRNet is gradually introduced. We begin by outlining our network in Section III-A, followed by the presentation of the proposed MGBF in Section III-B. Subsequently, the RAGR is discussed in Section III-C. The constructed HSCI was introduced in Section III-D. Finally, we conclude this section by presenting the comprehensive loss function.

A. Framework Overview

As depicted in Fig. 2, our proposed RAGRNet adheres to an encoder–decoder architectural framework, integrating a multiscale pyramid structure within the backbone network to capture diverse levels of information from the image. Concerning the encoder component, leveraging the potent feature extraction capabilities of Res2Net [67], we have selected it as our backbone. To align more effectively with ORSIs and tackle the challenging task of SOD, we have omitted the final adaptive average pooling layer and fully connected layer from the Res2Net [67] architecture. The remaining segments have been categorized into four distinct stages, namely, stage1, stage2, stage3, and stage4, resulting in the generation of feature maps f_e^1, f_e^2, f_e^3, and f_e^4, respectively. Subsequently, we have devised the MGBF, aimed at enhancing boundary information across varying scales, encompassing f_e^2, f_e^3, and f_e^4, thereby acquiring richer textural details. Following this, we have developed the RAGR, meticulously investigating the inherent associations between the semantically augmented feature f_e^4 and the finely-detailed boundary feature f_{MGBF} derived from the MGBF module. This process has led to the establishment of latent semantic correlations, yielding a feature representation that encapsulates comprehensive contextual relationships. Notably, our design of the graph reasoning module adopts a recurrent structure, wherein the output f_{RAGR} from the preceding graph reasoning module, along with f_e^4, is reintroduced into the graph reasoning module, thereby intensifying the internal relationships between salient objects. Finally, we have introduced a decoder named HSCI, harnessing the interplay between boundary and region information. This decoder employs combined 3×3 and 1×1 convolutions to serve as the saliency prediction head, culminating in the production of the definitive saliency detection output. Furthermore, without introducing supplementary information, we have utilized the Sobel operator to process the ORSIs, culminating in the creation of an authentic boundary supervisory map. Concurrently, throughout the training process, we have employed a diverse array of implicit supervisory cues, harmonizing with explicit supervision to ensure a comprehensive array of network constraints.

B. Multigranularity Boundary-Focused Module

Upon revisiting existing efforts in ORSI-SOD, it has come to our attention that the current endeavors fall short in their effective utilization of boundary information. Hence, we attempted to employ an MGBF to acquire edge information maps for subsequent processes. As shown in Fig. 3, the MGBF predominantly comprises inputs of multiscale information, culminating in the extraction of essential boundary features as its output. This strategic design not only minimizes the intricacies of later fusion procedures but also sharpens the module's focus and functionality. Meanwhile, it enhances semantic object modeling across distinct receptive fields. Moreover, bolstered by implicit supervisory signals, the MGBF facilitates the generation of more intricate boundary intricacies, consequently affording finer insights into boundaries.

In a formal sense, shallow layers have the capacity to capture intricate spatial details, such as texture features, while higher level features are better equipped to represent comprehensive semantic knowledge and identify salient objects. In a manner akin to the approaches taken by EMFINet [21]

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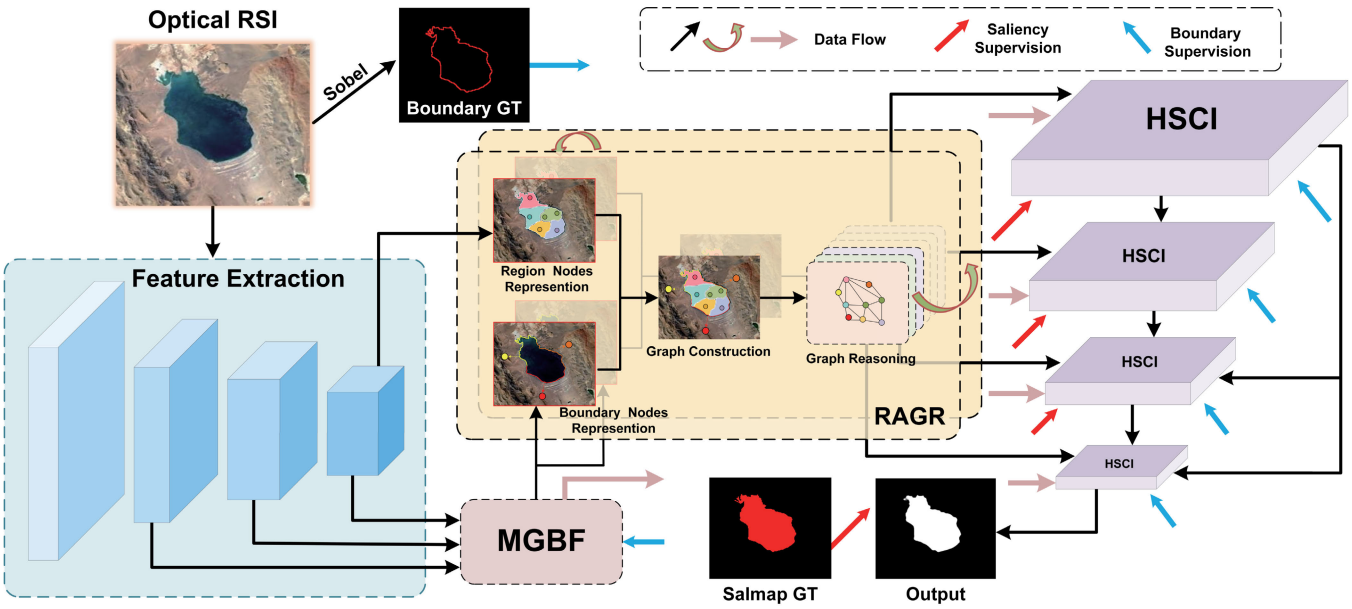


Fig. 2. Overview of the proposed RAGRNet. The framework leverages Res2Net as its backbone for feature extraction. Subsequently, the information flow is directed into the MGBF and RAGR, facilitating feature enhancement and relational inference. Finally, the enriched features are employed by the decoder, referred to as HSCI, to derive the saliency detection outcomes.

and EGNet [2], we merge the semantic information extracted from higher level networks with the finely-detailed boundary information derived from lower level networks. This fusion not only yields more credible cues regarding salient boundaries but also enables the network to synergistically learn across distinct tiers, enhancing the correlation between global and local contexts. The MGBF introduced in this study consists of three parallel branches, where f_e^2, f_e^3, and f_e^4 concurrently engage in meticulous boundary information processing. These branches are subsequently interlinked and fused to amalgamate insights from various semantic and boundary levels, thereby facilitating comprehensive comprehension of target structures. Given the irregular and diverse nature of objects in ORSIs, we endeavor to address this challenge by introducing deformable convolutions. Concretely, each branch commences with a 1×1 standard convolution and a 3×3 deformable convolution. The use of deformable convolutions allows for dynamic adjustments to the convolutional kernel shapes, offering a more precise matching to the topological characteristics of objects during the extraction of salient features. This adaptation enhances detection accuracy, aligning with the central theme of our work and the subsequent series of proposed components. Moreover, this configuration serves a dual purpose: first, the successful introduction of 1×1 convolution has facilitated the reduction of dimensionality along the channel axis, effectively alleviating computational burdens and model complexity. This operation enables linear combinations between channels, allowing the acquisition of more intricate feature representations. Second, the utilization of 3×3 convolution has not only optimized parameter quantities but has also augmented the local receptive field $\tilde{f}_e^2$, $\tilde{f}_e^3$, and $\tilde{f}_e^4$, achieving a balance between network parameterization and performance outcomes. Subsequent to this, to facilitate feature alignment and fusion, we utilize upsampling to expand $\tilde{f}_e^4$, enhancing its informational capacity to match that of $\tilde{f}_e^3$. Subsequently, these two are concatenated and dimensionally reduced through a 1×1 standard convolution. The process of leveraging deformable convolutions continues in parallel, extracting more elaborate irregular information from $\hat{f}_e^3$. Finally, mirroring the previous step, we expand $\hat{f}_e^3$ to align features with $\tilde{f}_e^2$, accompanied by 1×1 convolutions and batch normalization (BN) to render smoother features and enhance data robustness. This sequence of processes can be succinctly summarized as follows:

$$\tilde{f}_e^4 = \varpi\left(\xi\left(\psi\left(f_e^4\right)\right)\right) \tag{1}$$

$$\hat{f}_e^3 = \varpi\left(\xi\left(\psi\left(\gamma\left(\tilde{f}_e^4\right) + \tilde{f}_e^3\right)\right)\right) \tag{2}$$

$$f_{\mathrm{MGBF}} = \varpi\left(\xi\left(\psi\left(\gamma\left(\hat{f}_e^3\right) + \tilde{f}_e^2\right)\right)\right) \tag{3}$$

where ψ denotes a conventional 1×1 convolution, ξ represents a 3×3 deformable convolution with a 2-D kernel, ϖ signifies BN, γ corresponds to linear interpolation-based upsampling, and f_{MGBF} stands for the output generated by the MGBF.

C. RAGR Module

In ORSIs, artificial objects often exhibit topological structures and varying scales, which pose challenges of boundary ambiguity and region confusion in ORSI-SOD. In fact, the inherent interregion correlations play a crucial role in achieving fine-grained pixel-level segmentation. To address this, we propose a RAGR that progressively interacts between regional and boundary features to exploit their latent semantic complementarity, thereby effectively modeling the topological structures of objects.

Differing from the existing methods in ORSI-SOD like RRNet [20] and UG2L [66], we introduce boundary information and achieve deep fusion of regions and boundaries through adaptive feature reconstruction. As illustrated in Fig. 4,

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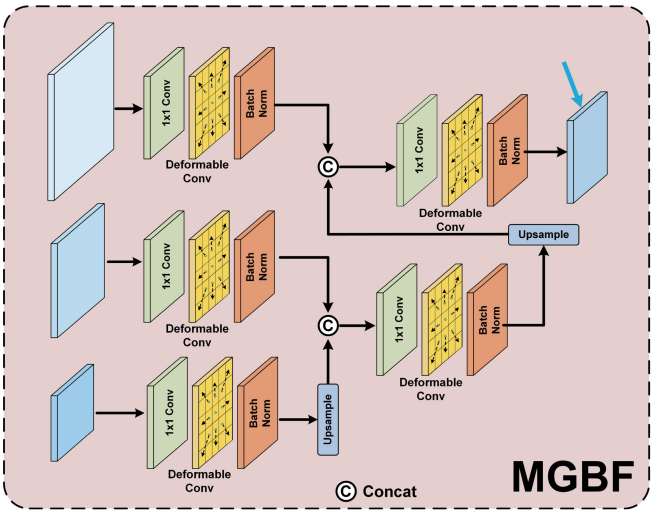


Fig. 3. Illustration of MGBF in RAGRNet.

we establish a recurrent projection-propagation-reprojection framework to realize structured multisource feature fusion and spatial relationship modeling. Specifically, given the representations of regional feature maps and boundary feature maps, we first project the regional features based on boundary features to construct a topological structure with fused boundary information. Subsequently, GCNs are employed for feature propagation to learn internode correlations. Finally, the graph representation is reprojected back to the pixel space and fused with the original regional features to refine target details. This cyclic mechanism fully leverages the complementarity between boundary and regional features, capturing object representations with correlated semantics, and significantly enhancing the detection performance of complex salient objects in remote sensing images. In the following, we will provide a detailed elaboration of this process.

1) Structured Projection: To facilitate the construction of a structured graph projection, the alignment of boundary features f_{MGBF} and regional features $F \in \mathbb{R}^{H\times W\times C}$ is initiated by bilinear interpolation to ensure commensurate spatial dimensions. Subsequently, the Hadamard product of the two is computed, assigning elevated weightage to contour pixels and thereby projecting the boundary cues into the regional feature space. This process culminates in the formulation of the graph $G = \{V, E\}$, where $V = V_R \cup V_B$ signifies the amalgamation of nodes from both regions and boundaries. In addition, anchoring points for the vertices are established via adaptive average pooling $A(\cdot)$ with a stride of s, employing a pooling kernel size of 6×6. These anchor points epitomize the centroids of subregions within the image. Subsequently, the similarity between regional features $R(F)$ and each anchor point is gauged and subjected to softmax normalization. Eventually, the resultant projection matrix A can be concisely expressed as

$$A = \mathrm{softmax}\left(A(R(F)\Theta f_{\mathrm{MGBF}}) \cdot R(F)^{\mathrm{T}}\right). \tag{4}$$

Here, Θ represents the Hadamard product. Simultaneously, the mapping of regional feature pixels onto the F can be expressed as follows:

$$F_G = A\psi(F) \tag{5}$$

where F_G encompasses the aggregation of cross-layer (both deep and shallow) and cross-domain (both region and boundary) messages from all adjacent nodes. This projection process achieves the aggregation of similar feature pixels onto the same node. Thus, each vertex symbolizes a specific subregion of the image, naturally encompassing not only node V_i^1 itself but also all its neighboring nodes represented by node V_i^{m-1}. Furthermore, both types of information can be aggregated into other connected nodes during the message-passing process.

2) Feature Relation Propagation: It is achieved by projecting boundary-guided features, thereby enhancing the relationships between nodes. Specifically, the resulting projection is combined with the projection derived from the region, not only effectively establishing strong associations between similar pixels and each individual regional node, but also playing a crucial role in facilitating the bidirectional flow of information between local and global contexts, thus enabling comprehensive information aggregation. In the subsequent step, the aggregated cross-layer (both deep and shallow) and cross-domain (both region and boundary) messages, represented as F_G, are introduced into the GCN. The method known as GCN represents an innovative approach for processing graph-structured data, distinguishing itself from traditional methods like CNN and transformer. The loading of the extracted embedding information F_G and the adjacency matrix A from (4) into GCN facilitates feature aggregation and updates until reaching the final equilibrium state. This operation effectively captures relationships between nodes, enhancing the model's capacity to discern intricate internode dependencies. In this process, specifically, F_G undergoes a transposition operation followed by a 1×1 convolutional operation. Moreover, the utilization of residual connections serves to update the node embedding, resulting in the generation of $F_G{}''$. This sequential process is succinctly summarized as

$$F_G{}' = \left[\psi\left(F_G{}^{\mathrm{T}}\right)\right]^{\mathrm{T}} - F_G \tag{6}$$

$$F_G{}'' = \mathrm{RELU}\left(\psi\left(F_G{}'\right)\right). \tag{7}$$

Here, RELU$(\cdot)$ as a nonlinear function.

3) Graph Representation Reprojection: To seamlessly integrate more correlated graph information into the existing model features, we opt to remap it back to the pixel coordinate space. However, since A is not necessarily a square matrix, its calculation can be relatively intricate. To address this, we consider the transpose matrix of A, denoted as A^{T}, as our reprojection matrix. Following the reprojection, a 1×1 convolution operation $\psi(\cdot)$ is applied to the features F to restore their dimensions, aligning them with the original feature channels. Subsequently, the sum of the original region feature map and the refined features postreprojection constitutes our preliminary output imbued with graph associations. Therefore, the corresponding pixel-level feature map, denoted as F_P, is defined as

$$F_P = F + \psi\left(A^{\mathrm{T}} F_G{}''\right). \tag{8}$$

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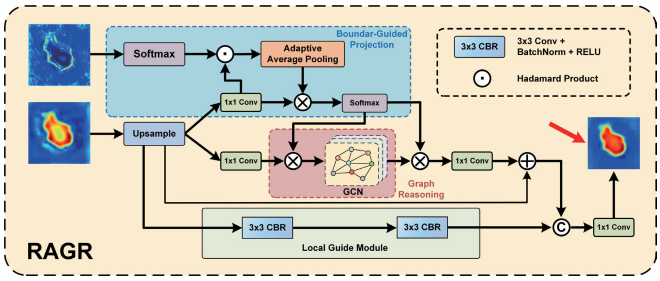


Fig. 4. Illustration of RAGR in RAGRNet.

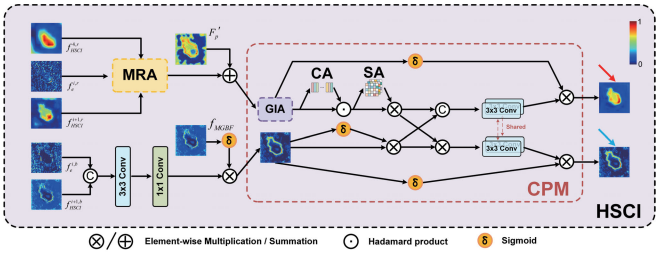


Fig. 5. Illustration of HSCI in RAGRNet.

Furthermore, compared to CNNs, graph models exhibit limitations in encoding local features. To address this deficiency and enhance the modeling of regional information, we concurrently introduce two sets of residual convolutional operations for input feature preprocessing in parallel with graph reasoning, named the local-guide model (LG). Each residual convolutional unit comprises a 3×3 deformable convolutional layer, followed by a BN layer and a ReLU activation function. This multilayer convolutional structure is adept at capturing elementary visual patterns, thereby furnishing ample local semantic priors for subsequent graph learning. Concurrently, the residual connections facilitate direct information propagation and reconstruction, mitigating semantic loss when stacking network layers. Following the convolutional operations, we attain an augmented regional feature representation denoted as f_{lg}. Subsequently, f_{lg} is combined with the output F_P and serves as the input, culminating in a contextually enriched feature representation $F_P{}'$ grounded in relationships, encapsulating both local details and global dependencies. The aforementioned process can be formalized as

$$F_P{}' = F_P + f_{\text{lg}}. \tag{9}$$

By integrating convolutional networks and graph models in this manner, the module is capable of simultaneously addressing the extraction of local details and the modeling of non-Euclidean spatial structures.

D. HSCI Decoder

To explore the underlying semantic complementarity between boundary features and region features more extensively, with the aim of effectively fusing diverse information sources, we have developed an innovative decoder named HSCI. Anchored in multitiered boundary and region features and empowered by the integration of graph models, this module achieves profound fusion and distillation of information. As depicted in Fig. 5, this module encompasses three key segments: a multilevel region feature aggregation, a multilevel boundary feature aggregation, and a cross-attention progressive module (CPM). The region and boundary branches progressively learn semantic traits, subsequently engaging in mutual enhancement and refinement through the cross-attention mechanism. This dynamic interplay incrementally refines salient object regions by iteratively honing distinct decoding insights.

1) Multilevel Region Perception Aggregation: To achieve enhanced aggregation of multilevel region information, we have devised the multilevel region aggregation (MRA) module for this purpose, as shown in Fig. 6. Initially, we perform element-wise fusion between outputs $f_e^{i,r}$ from the same-level encoder and the previous decoding layer information $f_{\text{HSCI}}^{i+1,r}$, this is done through element-wise multiplication to accentuate correlated features. Simultaneously, a residual connection with the original input is established by addition, thus amplifying feature representation. Subsequently, 3×3 convolutions are employed to extract associated information, which is then added to the point-wise fused features. Finally, a 1×1 convolution is applied to achieve feature fusion and obtain the processed output $f_e^{i,r'}$. The described procedure can be formalized as follows:

$$\breve{f}_e^{i,r} = \zeta\left(f_e^{i,r} + f_e^i \Theta f_{\text{HSCI}}^{i+1,r}\right) \tag{10}$$

$$\overleftrightarrow{f}_{\text{HSCI}}^{i,r} = \zeta\left(f_{\text{HSCI}}^{i+1,r} + f_e^{i,r} \Theta f_{\text{HSCI}}^{i+1,r}\right) \tag{11}$$

$$f_e^{i,r'} = \psi\left(\wp\left(\overleftrightarrow{f}_{\text{HSCI}}^{i,r}, \breve{f}_e^{i,r}\right)\right) \tag{12}$$

where $\wp$ denotes the concat operation. Subsequently, the introduction of $f_{\text{HSCI}}^{4,r}$, possessing richer semantic information, is performed. It is multiplied with $f_s^{i,r'}$ to enhance the feature representation. This enhanced representation is then appended to $f_{\text{HSCI}}^{4,r}$ to achieve feature enrichment, followed by a feature smoothing operation. In order to better capture intricate feature patterns and suppress noise effectively, we have incorporated GELU operations and integrated channel attention (CA). This sequence of steps can be expressed as follows:

$$f_e^{i,r''} = \eta\left(\mu\left(\psi\left(f_{\text{HSCI}}^{4,r} + f_{\text{HSCI}}^{4,r} \Theta f_e^{i,r'}\right)\right)\right) \tag{13}$$

where μ represents the GELU, and η denotes CA. Subsequently, the combination with $F_P{}'$ from the graph reasoning network results in $f_s^{i,r''}$, introducing structural awareness to provide multiscale semantic information for the regional data, thereby enhancing the network's sensitivity to shape variations.

2) Multilevel Boundary Perception Aggregation: Similar to the multilevel region perception aggregation, we adopt a multilevel information fusion approach to reprocess the prominent boundary information. Initially, $f_e^{i,b}$ from the same-level encoder and $f_{\text{HSCI}}^{i+1,b}$ from the previous decoder are concatenated. Subsequently, a 3×3 convolution is applied, followed by BN and RELU activation to refine the information, resulting in $f_e^{i,b'}$. Furthermore, a multiplication and addition operation

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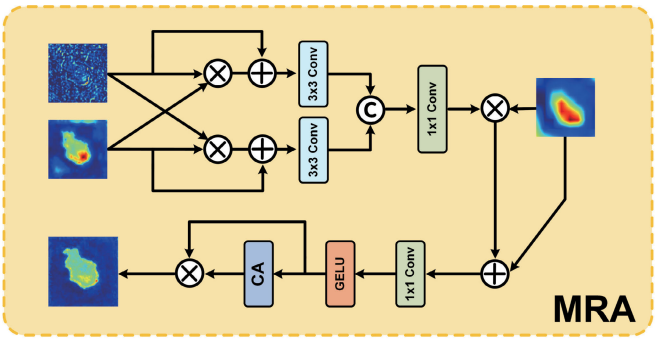


Fig. 6. Illustration of MRA in HSCI.

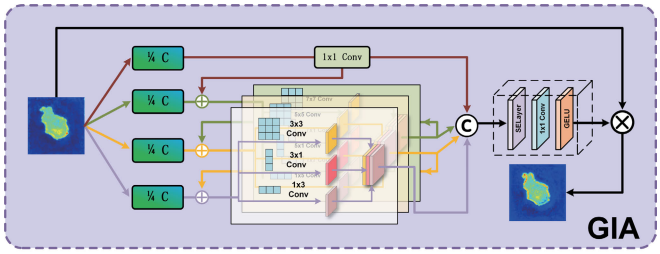


Fig. 7. Illustration of GIA in HSCI.

with f_{MGBF} is conducted to emphasize significant details, leading to $f_e^{i,b''}$. This process can be represented as follows:

$$f_e^{i,b'} = \psi\left(\vartheta\left(\wp\left(f_{\mathrm{HSCI}}^{i+1,b}, f_e^{i,b}\right)\right)\right) \tag{14}$$

$$f_e^{i,b''} = f_e^{i,b'} \odot \delta(f_{\mathrm{MGBF}}) + f_e^{i,b'} \tag{15}$$

where ϑ denotes the 3×3 Conv, BN, and RELU, and δ represents the sigmoid operation.

3) Cross-Attention Progressive Module: Region features encompass semantic aspects such as object classification, while boundary features offer structural information like object scale and shape. To fully harness their complementary nature, we devised parallel subbranches for regions and boundaries. Given the intricate nature of remote sensing scenes, and to enhance the network's ability to perceive multiscale semantic structures, we incorporated the grouped-progressive imbalanced-context aggregation module (GIA), as shown in Fig. 7, into the region subbranch. Specifically, the input features are initially segmented into groups and linearly mapped to acquire subsets of subspace representations that are channel-adjusted

$$P = P_1, P_2, P_3, P_4, P_i \in \mathbb{R}^{C\times H\times W} \tag{16}$$

$$P_i' = w_i P_i, w_i \in \mathbb{R}^{C_i\times C_i}, \ i = 1, 3, 5, 7. \tag{17}$$

Here, P_i represents the split channel of the ith group, w_i denotes the convolution weights of various kernel sizes, and P_i' signifies the mapped features. Subsequently, a distinct four-stage asymmetric convolutional structure was constructed to progressively expand the receptive field. The first group comprises $\{1 \times 1\}$, the second group includes $\{3 \times 1, 3 \times 3, 1 \times 3\}$, the third group encompasses $\{5 \times 1, 5 \times 5, 1 \times 5\}$, and the fourth group consists of $\{7 \times 1, 7 \times 7, 1 \times 7\}$ kernels. The asymmetric convolution kernel groups at each stage aggregate context dependencies from different ranges directionally, drawing inspiration from the residual concept. These kernels are concatenated with the previous stage's subspace representation, guiding each stage's feature learning with a broader context. With the advancement of stages, the network's field of view progressively expands, allowing the feature subspace to absorb abundant multiscale contextual information

$$F_{\mathrm{GIA}}^1 = w_{1\times 1} * P_1', \ i = 1 \tag{18}$$

$$F_{\mathrm{GIA}}^i = \left[\mathcal{F}_i\left(\mathrm{Concat}\left(P_i', F_{\mathrm{GIA}}^{i-1}\right)\right]\right.$$

$$= \left[(w_{2i-1\times 1}, w_{2i-1\times 2i-1}, w_{1\times 2i-1}) \left(\mathrm{Concat}\left(P_i', F_{\mathrm{GIA}}^{i-1}\right)\right]\right., \ i \in 2, 3, 4. \tag{19}$$

Here, $w_{2i-1\times 1}$, $w_{2i-1\times 2i-1}$, and $w_{1\times 2i-1}$ represent convolutional weights for distinct kernel sizes, and * denotes the convolution operation. $\mathcal{F}_i$ signifies the nonuniform convolution operation for the ith group, while F_{GIA}^i corresponds to the output of the respective layer. To cater to the interplay between feature channels and spatial characteristics in intricate scenarios, we introduce a dual-attention mechanism, encompassing CA and spatial attention (SA). This yields $F_{\mathrm{GIA}}{}'$, thereby converging diverse tiers of contextual relationships. On the whole, this module empowers the network with the ability to interpret the semantic intricacies of complex scenes while also efficiently managing parameter intricacies.

Considering the significance of boundary representation in detection tasks, an activation modulation mechanism has been introduced within the boundary information branch for $f_e^{i,b''}$ to enhance the refinement of boundary features. Specifically, $f_e^{i,b''}$ undergoes a sigmoid activation, resulting in the probabilistic boundary response $\tilde{f}_e^{i,b''}$, which amplifies the clarity and distinctiveness of the boundaries. To retain the original boundary intricacies, $f_e^{i,b''}$ and $\tilde{f}_e^{i,b''}$ are subjected to a Hadamard multiplication, involving element-wise multiplication, yielding the modulated boundary feature $f_e^{i,b'''}$.

In the end, we propose a multipath feature fusion approach to leverage complementary information. We merge region feature $F_{\mathrm{GIA}}{}'$ and boundary feature $f_e^{i,b'''}$ through concatenation and Hadamard multiplication. These pathways then undergo parallel processing with 3×3 convolution layers for spatial abstraction. Activation modulation and feature selection guide the feature learning process, enhancing sensitivity for target detection. Optimized pathway features contribute to region representation and boundary delineation, harmonizing explicit and implicit cues for directed supervision and optimization.

E. Loss Function

To achieve synergistic optimization of region and boundary features, we introduce a multitask joint supervision loss function, drawing inspiration from [68], [69], and [70]. Specifically, we commence by supervising the multiscale semantic segmentation results and boundary results outputted by the HSCI. For the semantic segmentation task, we employ the binary cross entropy (BCE) loss, intersection over union (IoU) loss, and focal loss (Floss) to comprehensively assess the disparities between the predicted semantic segmentation map

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and the corresponding ground truth (GT). These three loss functions impose multifaceted constraints and regulations on the semantic segmentation outcomes. Regarding boundary prediction results, we adopt the BCE loss function for supervision, given the relatively straightforward nature of the binary classification involved in boundary prediction. Formally, the formulated multitask loss function is as follows:

$$\mathcal{L}_{\text{seg}} = \sum_{i=1}^{4} \left[\tau L^i_{\text{BCE}}(S_t, L_t) + \sigma L^i_{\text{IoU}}(S_t, L_t) + \lambda L^i_{\text{Floss}}(S_t, L_t) \right] \tag{20}$$

$$\mathcal{L}_{\text{edge}} = \sum_{i=1}^{4} \left[\varepsilon L^i_{\text{BCE}}(b_t, B_t) \right] \tag{21}$$

$$\mathcal{L}_{\text{decode}} = \mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{edge}} \tag{22}$$

where the variables S_t and b_t represent the outputs related to regions and boundaries from each decoder, respectively. The variables L_t and B_t indicate the GT pertaining to regions and boundaries. The loss functions L_{BCE}, L_{IoU}, and L_{Floss} correspond to the BCE, IoU, and Floss, respectively. Their respective contributions to the overall network loss are determined by the factors τ, σ, and λ. In this context, τ, σ, and λ have been uniformly assigned a value of 1. The terms $\mathcal{L}_{\text{seg}}$ and $\mathcal{L}_{\text{edge}}$ refer to the semantic segmentation and boundary prediction loss of the decoder. In addition, $\mathcal{L}_{\text{decode}}$ signifies the cumulative loss value for the decoder.

In order to steer the network toward a focused learning of boundary representation, we introduce the BCE loss to the newly generated boundary features. This ensures a continual minimization of the error between the predicted boundary image and its corresponding annotated boundary image. Distinguishing from the direct supervision of initial boundaries, the boundary feature aggregation module, through feature transformation, reinforces a more abstract and refined boundary representation

$$\mathcal{L}_{\text{edge}}{}' = \varepsilon L_{\text{BCE}}\left(\tilde{b}_t, B_t\right). \tag{23}$$

Here, $\tilde{b}_t$ represents the new boundaries generated by the boundary aggregation module, while $\mathcal{L}_{\text{edge}}{}'$ denotes the boundary loss supervised by the boundary aggregation module. The ε signifies the contribution of L_{BCE} to the overall loss function, and it has been set to a value of 1 as per our definition. Finally, to compel the RAGR to learn more accurate feature information in modeling object-aware relational patterns, we apply deep supervision to its outputs. This involves the use of BCE loss, IoU loss, and Floss to comprehensively assess the discrepancies between the predicted semantic segmentation maps and the GT results. This approach enhances the module's capacity to capture precise object-related relationships

$$\mathcal{L}_{\text{graph}} = \sum_{i=1}^{2} \mathcal{L}^i_{\text{graph}} = \sum_{i=1}^{2} \left[\tau L^i_{\text{BCE}}(r_t, L_t) + \sigma L^i_{\text{IoU}}(r_t, L_t) + \lambda L^i_{\text{Floss}}(r_t, L_t) \right]. \tag{24}$$

Here, r_t signifies the region feature information generated by the first and second graph reasoning modules, while $\mathcal{L}_{\text{graph}}$ represents the loss value of the RAGR module.

In summary, the multitask loss function can be expressed as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{decode}} + \mathcal{L}_{\text{edge}}{}' + \mathcal{L}_{\text{graph}} \tag{25}$$

where $\mathcal{L}_{\text{total}}$ represents the overall loss of the network, which is adjusted through an ongoing positive feedback loop to optimize the network's learning capabilities. Finally, we elucidate the inference details, meticulously delineated in Algorithm 1.

Algorithm 1 : The inference details of proposed RAGRNet

Input: Optical remote sensing images set $\{I_1, I_2, I_3 \cdots I_n\} \in \mathbb{R}^{3\times256\times256}$, Label (Ground Truth) set $\{L_1, L_2, L_3 \cdots L_n\} \in \mathbb{R}^{1\times256\times256}$

Output: Salmap $Sal \in \mathbb{R}^{1\times256\times256}$

while n in epoch N **do**
- Sample I_i, L_i from $\{I_1, I_2, I_3 \cdots I_n\}$ and $\{L_1, L_2, L_3 \cdots L_n\}$
- Step1: Get boundary map $\{B_1, B_2, B_3 \cdots B_n\} \in \mathbb{R}^{1\times256\times256}$ from GT through Sobel operator
- Step2: Acquire Mutli-level feature from Backbone
- $f_e^1 \cdots f_e^4 \leftarrow backbone(I_i)$
- Step3: Put f_e^2, f_e^3, f_e^4 into MGBF to make comprehensive fine boundary information f_{MGBF}
- $f_{\text{MGBF}} \leftarrow MGBF(f_e^2, f_e^3, f_e^4)$
- Step4: Modeling boundaries f_{MGBF} and regional F relationships
- // Structured Projection
- Calculate the projection matrix A of sampled graph
- $A \leftarrow \text{softmax}(A(R(F)\Theta f_{\text{MGBF}}) \cdot R(F)^{\text{T}})$
- Aggregation of similar feature pixels onto the same node
- $F_G \leftarrow A\psi(F)$
- // Feature Relation Propagation
- $F_G{}'' \leftarrow \text{RELU}(\psi(F_G{}'))$, $F_G{}' = [\psi(F_G{}^{\text{T}})]^{\text{T}} - F_G$
- // Graph Representation Reprojection
- remap back to the pixel coordinate space
- $F_P \leftarrow F + \psi(A^{\text{T}} F_G{}'')$
- // Modeling of non-Euclidean spatial structures and local feature
- $F_P{}' \leftarrow F_P + f_{\text{lg}}$
- Step5: Heterogeneous semantic cross-progressive information decoding
- $S, b \leftarrow CPM[MRA(f_{\text{HSCI}}^{4,r}, f_e^{i,r}, f_{\text{HSCI}}^{i+1,r}) + F_P{}', f_e^{i,b}, f_{\text{HSCI}}^{i+1,b}, f_{\text{MGBF}}]$
- Compute the loss to update the RAGRNet
- $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{decode}} + \mathcal{L}_{\text{edge}}{}' + \mathcal{L}_{\text{graph}}$
- Return Sal

end while

IV. Experimental Analysis

In this section, we commence by outlining the experimental configuration, followed by presenting various quantitative and qualitative analyses, as well as performing ablation experiments. Subsequently, an analysis of instances of failure encountered is undertaken. Finally, we illustrate the generalizability to large-area detection.

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A. Datasets and Experimental Setting

1) Datasets: To thoroughly evaluate the robustness of our proposed model, we conducted extensive experiments using four challenging optical remote sensing datasets. The initial dataset, ORSSD [12], was the pioneering release for ORSI-SOD, comprising 600 training samples and 200 test samples. An extended version, EORSSD [19], augmented the dataset to 2000 samples, including 1400 for training and the rest for testing. EORSSD [19] introduced more intricate scenarios with multiscale objects, compared to the original ORSSD. The largest dataset among the ORSI-SOD tasks, ORSI-4199 [46], featured 4199 images, split into 2000 for training and 2199 for testing. Unlike the preceding datasets, ORSI-4199 [46] encompassed complex remote sensing images, encompassing diverse scenes like storage oil tanks, rural buildings, and urban landmarks, categorized into nine distinct scenes. UGSet [53] is a comprehensive dataset tailored for extensive UGS mapping, constructed with a large scale and high resolution. It encompasses 4544 multiscene images, each with a spatial resolution of approximately 1 m and dimensions of 512 × 512 pixels. The dataset is systematically divided into training, validation, and test sets with a distribution ratio of 5:2:3. To align with the previously mentioned dataset, we have restructured UGSet, allocating 3631 images for the training and 913 images for the validation, maintaining an 8:2 proportion. All datasets are accompanied by pixel-level annotations conducted by expert visual interpretation, expediting the training and testing of the network. Similar to [15], [44], and [71], data augmentation was applied to all datasets, further heightening the network's robustness.

2) Evaluation Criteria: To ensure a comprehensive and fair multidimensional comparison, we employed ten widely recognized quantitative metrics across the ORSSD, EORSSD, ORSI-4199, and UGSet. These metrics encompassed Precision–Recall (PR) curves, F-measure curves, mean absolute error (MAE, $\mathcal{M}$), S-measure (S_α), adaptive F-measure (F_β^{adp}), mean F-measure (F_β^{mean}), max F-measure ($F_\beta^{\max}$), adaptive E-measure (E_ξ^{adp}), mean E-measure (E_ξ^{mean}), and max E-measure ($E_\xi^{\max}$). Notably, the MAE metric follows a "lower is better" criterion. Conversely, for the F-measure, E-measure, and S-measure, the principle is "higher is better." The MAE quantifies the one-norm discrepancy between the generated saliency map and the GT, offering a precise gauge of the disparity between these two components

$$\text{MAE} = \frac{1}{H \times W} \sum_{x=1}^{H} \sum_{y=1}^{W} |S(x, y) - GT(x, y)| \tag{26}$$

where the variables H and W correspond to the height and width dimensions. The S-measure quantifies the structural similarity that encompasses both object-aware and region-aware aspects between the predictions and GTs. Mathematically, its formulation is given by

$$S_m = \alpha * S_o + (1 - \alpha) * S_r \tag{27}$$

where the coefficient α is assigned a value of 0.5 to ensure equal consideration of both similarities, as previously suggested [10]. Simultaneously, this coefficient is employed to strike a balance between object-aware similarity (S_o) and region-aware similarity (S_r). The F-measure, serving as an all-encompassing assessment, amalgamates Precision and Recall. Its mathematical expression is as follows:

$$F_\beta = \frac{\left(1 + \beta^2\right) \times \text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}}. \tag{28}$$

Here, the parameter β^2 is assigned a value of 0.3, aligning with the approach undertaken in [49], to underscore Precision's significance over Recall. The E-measure evaluates the similarity between the predicted saliency maps and the GT while considering both local and image-level saliency values simultaneously

$$E_\phi = \frac{1}{H \times W} \sum_{x=1}^{H} \sum_{y=1}^{W} \theta(\phi) \tag{29}$$

where ϕ represents the alignment matrix, while $\theta(\phi)$ signifies the refined alignment matrix as introduced in [72]. We employ various thresholds to discretize the saliency prediction images. Subsequently, PR pairs are generated from the binary outcomes and the GT. Following these procedures, PR and F-measure curves can be graphed. The PR curve approaching the coordinates (1, 1) indicates improved model performance. For ease of evaluation, we have adopted the assessment methodology proposed by Fan et al. [73].

In addition, three computational complexity metrics were incorporated to provide a comprehensive assessment of overall performance from an alternative perspective. These metrics encompass the inference speed measured in frames per second (FPS), the parameter count denoted as the number of parameters (Params) expressed in millions (M), and the volume of floating-point operations (FLOPs) measured in billions (G). While higher values are preferred for inference speed, the opposite holds true for the latter two metrics. The reported inference speed is derived with a batch size of 1 and excludes any I/O time.

3) Implementation Details: We have implemented the proposed RAGRNet network using the PyTorch framework on a Linux platform equipped with the NVIDIA V100 GPU. To expedite network performance and conserve time, we leveraged the pretrained weights of Res2Net [67], obtained through training on the ImageNet dataset, as part of our network initialization. This pretraining was conducted over 50 epochs. Following the approach of prior research [31], the input images were rescaled to multiple dimensions using scaling factors of [0.75, 1, 1.25] as a data augmentation technique. The RAGRNet was trained using the Adam [74] optimization algorithm. Throughout the training process, we set the batch size to 8 and initialized the learning rate to 5e-5. To enhance stability and convergence, we incorporated a learning rate decay strategy, adjusting the relevant parameter by a factor of 0.1 after 30 epochs. Furthermore, all samples in both the training and testing sets were uniformly resized to dimensions of 256 × 256.

4) Baselines: We conducted a comprehensive comparative analysis of our approach against 38 SOTA methods in the domains of NSI-SOD and ORSI-SOD. The evaluation was

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performed on three benchmark datasets, as outlined in Tables I and II. The considered methods encompass a spectrum of categories, including traditional NSI-SOD methods (such as RRWR [27], DSG [75], and VST [37]), CNN-based methods (comprising DSS [29], RADF [76], EGNet [2], PoolNet [30], GCPA [3], MINet [34], GateNet [77], U2Net [78], F3Net [10], SUCA [35], and PA-KRN [11]), traditional ORSI-SOD methods (encompassing VOS [71], CMC [44], SMFF [42], and HFANet [79]), CNN-based ORSI-SOD methods (including LVNet [12], DAFNet [19], EMFINet [21], SARNet [15], ERPNet [49], MCCNet [13], MSCNet [80], AGNet [81], RRNet [20], FSMINet [82], MJRBM [46], ACCoNet [48], SeaNet [17], CorrNet [16], BAFSNet [68], PCDNet [83], SDNet [84], and AESINet [69]), and transformer-based ORSI-SOD methods (ASTT [50] and BMGT [70]).

To ensure rigor, all reported results were generated by retraining the models using publicly available source code and default parameter settings. In cases where authors had shared their outcomes, we also included those results for reference. Notably, given the novelty of the ORSI-4199 dataset, some of the model outcomes for this dataset are still under refinement. Consequently, we opted to present a selection of representative results. In instances where certain performance metrics were not ascertainable, we indicated this using the "-" notation within the corresponding table entries.

B. Comparison With SOTA Methods

1) Quantitative Comparison: It presents a comprehensive comparison of our model with SOTA models using eight evaluation metrics in Table I. These models fall into five categories: three traditional NSI-SOD models, 11 CNN-based NSI-SOD models, four traditional ORSI-SOD models, 18 CNN-based ORSI-SOD models, and two transformer-based ORSI-SOD models. The analysis indicates that, in most cases, CNN-based approaches outperform traditional ORSI-SOD methods, with the exception of HFANet. Despite trailing slightly behind CNN-based ORSI-SOD models in specific aspects, RAGRNet, as a CNN-based method, maintains a leading position in terms of overall performance. Specifically, within the EORSSD dataset, RAGRNet ranks first in five metrics. While it is outperformed by HFANet, MCCNet, and DAFNet in S_α, $F_\beta^{\max}$, and $E_\xi^{\max}$, it significantly excels in F_β^{mean}, surpassing HFANet by 0.0089, MCCNet by 0.0925, and DAFNet by 0.0166. In comparison to BAFSNet, RAGRNet also demonstrates satisfactory outcomes, achieving $F_\beta^{\max}$ of 0.8852 (RAGRNet) compared to 0.8708 (BAFSNet), and $E_\xi^{\max}$ of 0.9785 (RAGRNet) compared to 0.9772 (BAFSNet). In the context of ORSSD, RAGRNet emerges as the leader across all eight metrics, showcasing its strong performance. For example, it outperforms AESINet by 0.0195 in F_β^{mean}, MCCNet by 0.0021 in $\mathcal{M}$, and BAFSNet by 0.0140 in F_β^{mean}, while also leading in S_α and F_β^{adp} by 0.0122. Notably, RAGRNet stands as the only model with an $\mathcal{M}$ value below 0.007, specifically at 0.0066, among all the evaluated models. To facilitate a more comprehensive comparison, we employed F-measure curves [depicted in Fig. 8(a) and (c)] and the PR curves [illustrated in Fig. 8(b) and (d)] to measure. In Fig. 8, it is evident that RAGRNet demonstrates a clear advantage on the ORSSD dataset. Specifically, on the PR curve, RAGRNet closely approaches (1, 1), and the F-measure curve approaches its peak, corroborating the data shown in Table I. However, on EORSSD, RAGRNet does exhibit occasional lapses but still maintains a second-place position. In summary, our method excels over competing methods in a comprehensive evaluation, underscoring its significant achievements in the field.

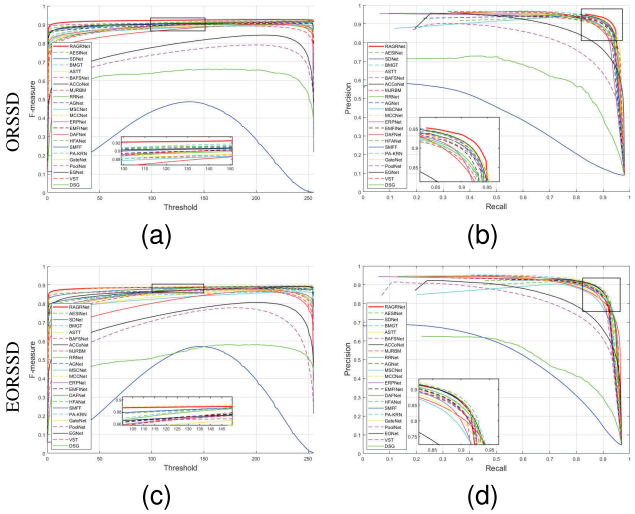


Fig. 8. (a) and (c) F-measure curve and (b) and (d) PR curve of different methods on the testing dataset of the ORSSD [12] and EORSSD [19]. Please zoomed-in view for the best view for color.

In order to delve further into the model's robustness, we conducted additional training on the ORSI-4199 [46] dataset using the same parameters. The outcomes of this retraining effort are outlined in Table II. The results are compelling, highlighting that our model secured the first rank in four distinct metrics, attained the second position in three metrics, and clinched the third position in one metric. When compared to ASTT, our model showcased noteworthy enhancements, particularly evident in E_ξ^{adp} with a gain of 0.0045, and F_β^{adp} with an impressive boost of 0.0231. Similarly, in comparison with ACCoNet, our model exhibited improvements of 0.005 in the S_α and 0.003 in the $\mathcal{M}$. Furthermore, in comparison to the traditional NSI-SOD transfer method VST, our model achieved significant advancements, with the most remarkable progress observed in E_ξ^{adp}, demonstrating an increase of 0.0264. These findings tangibly underscore the robust performance of our model, effectively affirming the merit of our approach in establishing meaningful inference relationships between boundaries and regions.

2) Qualitative Comparison: We present a series of compelling visual results in our article, showcasing the performance of RAGRNet in comparison to other models on a set of representative scenes depicted in Fig. 9. These scenes can be categorized into five groups: clusters of diminutive salient objects, objects in low-light conditions, objects within intricate cluttered backgrounds, irregular objects in commonplace scenarios, and challenging objects with extreme topologies.

Due to the aerial nature of remote sensing images captured by satellites, scenarios featuring small objects are

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TABLE I

QUANTITATIVE RESULTS ON EORSSD [19] AND ORSSD [12]. THERE ARE 38 SOTA METHODS CAN BE CLASSIFIED INTO THREE TRADITIONAL NSI-SOD (T.N.), 11 CNN-BASED NSI-SOD (C.N.), FOUR TRADITIONAL ORSI-SOD (T.R.), 18 CNN-BASED ORSI-SOD (C.R.), AND TWO TRANSFORMER-BASED ORSI-SOD (T.O.). ↑/↓ MEANS A LARGER/SMALLER SCORE IS BETTER. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE, RESPECTIVELY

Methods	Publication	Type	Speed (*fps*) ↑	#Params (M) ↓	FLOPs (G) ↓	EORSSD [19]								ORSSD [12]							
						$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
RRWR [27]	2015 CVPR	T.N.	0.3	-	-	.5992	.3993	.3686	.3344	.6894	.5943	.5639	.1677	.6835	.5590	.5125	.4874	.7649	.7017	.6949	.1324
DSG [75]	2017 TIP	T.N.	0.6	-	-	.6428	.5259	.4597	.4012	.7286	.6954	.6269	.1246	.7195	.6238	.5747	.5657	.7912	.7337	.7538	.1024
VST [37]	2021 ICCV	T.N.	23	44.03	23.2	.9203	.8724	.8257	.7015	.9736	.9417	.8872	.0067	.9318	.8999	.8710	.8126	.9755	.9596	.9417	.0100
DSS [29]	2017 CVPR	C.N.	8	62.23	114.6	.7874	.6868	.5801	.4597	.9205	.7631	.6933	.0186	.8262	.7467	.6962	.6206	.8860	.8362	.8085	.0363
RADF [76]	2018 AAAI	C.N.	7	62.54	214.2	.8188	.7473	.6582	.4933	.9162	.8567	.7160	.0168	.8259	.7619	.6856	.5730	.9130	.8298	.7678	.0382
EGNet [2]	2019 ICCV	C.N.	9	108.07	159	.8601	.8453	.6967	.5379	.9570	.8775	.7566	.0110	.8774	.8492	.7500	.7415	.9731	.9013	.8851	.0216
PoolNet [30]	2019 CVPR	C.N.	25	53.63	49	.8869	.8347	.7905	.6723	.9524	.9167	.8647	.0102	.9026	.8687	.8433	.7861	.9509	.9341	.9205	.0168
MINet [34]	2020 CVPR	C.N.	12	47.56	146.3	.9040	.8344	.8174	.7705	.9442	.9346	.9243	.0093	.9040	.8761	.8574	.8251	.9545	.9454	.9423	.0144
GCPA [3]	2020 AAAI	C.N.	23	67.06	54.3	.8869	.8347	.7905	.6723	.9524	.9167	.8647	.0102	.9026	.8687	.8433	.7861	.9509	.9341	.9205	.0168
GateNet [77]	2020 ECCV	C.N.	25	100.02	108.3	.9114	.8566	.8228	.7109	.9610	.9385	.8909	.0095	.9186	.8913	.8679	.8229	.9719	.9562	.9428	.0131
U2Net [78]	2020 PR	C.N.	25	44.01	37	.9199	.8732	.8329	.7221	.9649	.9373	.8989	.0076	.9162	.8738	.8492	.8038	.9539	.9387	.9326	.0166
F3Net [10]	2020 AAAI	C.N.	-	25	8	.9020	.8358	.7924	.7330	.9573	.9421	.9153	.0089	.9170	.8762	.8526	.8164	.9651	.9559	.9441	.0152
SUCA [35]	2020 TMM	C.N.	24	117.71	56.4	.8998	.8229	.7949	.7260	.9520	.9277	.9082	.0097	.8989	.8484	.8237	.7748	.9584	.9400	.9194	.0145
PA-KRN [11]	2021 AAAI	C.N.	16	141.06	617.7	.9192	.8639	.8358	.7993	.9619	.9536	.9416	.0104	.9239	.8890	.8727	.8548	.9680	.9620	.9579	.0139
VOS [71]	2018 GRSL	T.R.	-	-	-	.5082	.2765	.2107	.1836	.5982	.4886	.4767	.2096	.5366	.3471	.2717	.2633	.6514	.5352	.5826	.2151
CMC [44]	2019 RS	T.R.	-	-	-	.5798	.3268	.2692	.2007	.6803	.5894	.4890	.1057	.6033	.3913	.3454	.3108	.7064	.6417	.5996	.1267
SMFF [42]	2019 IJRS	T.R.	-	-	-	.5401	.5176	.2992	.2083	.7744	.5197	.5014	.1434	.5312	.4417	.2684	.2496	.7402	.4920	.5676	.1854
HFANet [79]	2022 TGRS	T.R.	26	60.53	68.3	.9380	.8876	.8681	.8365	.9740	.9679	.9644	.0070	.9399	.9112	.8981	.8819	.9770	.9712	.9722	.0092
LVNet [12]	2019 TGRS	C.R.	1.4	207	-	.8644	.8051	.7328	.6306	.9279	.8801	.8465	.0145	.8815	.8414	.7995	.7506	.9456	.9259	.9195	.0207
SARNet [15]	2021 RS	C.R.	47	25.91	129.7	.9240	.8719	.8541	.8304	.9620	.9555	.9536	.0099	.9134	.8850	.8619	.8512	.9557	.9477	.9464	.0187
DAFNet [19]	2022 TIP	C.R.	26	29.35	68.5	.9166	.8614	.7845	.6427	.9861	.9291	.8446	.0060	.9191	.8928	.8511	.7876	.9771	.9539	.9360	.0113
EMFINet [21]	2022 TGRS	C.R.	25	107.26	480.9	.9290	.8720	.8486	.7984	.9711	.9604	.9501	.0084	.9366	.9002	.8856	.8617	.9737	.9671	.9663	.0109
ERPNet [49]	2022 TCYB	C.R.	50	56.48	87.2	.9210	.8632	.8304	.7554	.9603	.9401	.9228	.0089	.9254	.8974	.8745	.8356	.9710	.9566	.9520	.0135
MCCNet [13]	2022 TGRS	C.R.	95	67.65	117	.9327	.8904	.8604	.8137	.9755	.9685	.9538	.0066	.9437	.9155	.9054	.8957	.9800	.9758	.9735	.0087
MSCNet [80]	2022 ICPR	C.R.	55	3.29	23.38	.9086	.8539	.8151	.7553	.9689	.9551	.9329	.0090	.9227	.9107	.8676	.8350	.9754	.9653	.9584	.0129
AGNet [81]	2022 ICANN	C.R.	-	26.60	9.65	.9287	.8785	.8516	.8021	.9752	.9656	.9540	.0067	.9389	.9098	.8956	.8723	.9811	.9728	.9729	.0091
RRNet [20]	2022 TGRS	C.R.	41.5	68	190	.9266	.8781	.8351	.7251	.9716	.9415	.9034	.0075	.9260	.8923	.8667	.8162	.9687	.9498	.9434	.0114
FSMINet [82]	2022 GRSL	C.R.	17.7	3.6	9.7	.9255	.8678	.8436	.8015	.9666	.9567	.9490	.0079	.9361	.9041	.8878	.8710	.9759	.9672	.9693	.0101
MJRBM [46]	2022 TGRS	C.R.	32	43.54	95.7	.9197	.8701	.8239	.7066	.9646	.9350	.8897	.0099	.9204	.9025	.8566	.8022	.9623	.9415	.9328	.0145
ACCoNet [48]	2023 TCYB	C.R.	81	102.55	368.6	.9290	.8837	.8552	.8008	.9727	.9653	.9454	.0074	.9437	.9149	.8971	.8806	.9796	.9754	.9721	.0088
SeaNet [17]	2023 TGRS	C.R.	96	2.76	1.7	.9208	.8649	.8519	.8304	.9710	.9651	.9602	.0073	.9260	.8942	.8772	.8625	.9767	.9722	.9670	.0105
CorrNet [16]	2023 TGRS	C.R.	100	4.09	21.1	.9289	.8778	.8620	.8311	.9696	.9646	.9593	.0083	.9380	.9129	.9002	.8875	.9790	.9746	.9721	.0098
BAFSNet [68]	2023 TGRS	C.R.	75	31	16	.9286	.8708	.8583	.8477	.9772	.9735	.9729	.0060	.9385	.9136	.9017	.8982	.9819	.9785	.9779	.0079
PCDNet [83]	2023 TGRS	C.R.	30.2	17.7	52.9	.9346	.8830	.8590	.8088	.9744	.9645	.9507	.0069	.9444	.9138	.9014	.8839	.9809	.9742	.9731	.0094
SDNet [84]	2023 TGRS	C.R.	-	-	-	.9359	.8853	.8680	.8370	.9704	.9644	.9607	.0065	.9340	.9074	.8932	.8752	.9768	.9717	.9677	.0101
AESINet [69]	2023 TGRS	C.R.	23	-	-	.9347	.8792	.8496	.7909	.9757	.9647	.9480	.0064	.9455	.9160	.8962	.8640	.9814	.9741	.9689	.0085
ASTT [50]	2023 TGRS	T.O.	13	44.6	69.45	.9253	.8741	.8297	.7534	.9757	.9584	.9247	.0059	.9348	.9061	.8811	.8456	.9795	.9699	.9587	.0094
BMGT [70]	2023 IJRS	T.O.	35	3.8	22.7	.9292	.8829	.8409	.7245	.9740	.9484	.8972	.0058	.9368	.9105	.8871	.8370	.9781	.9610	.9529	.0092
Ours	-	C.R.	31	35.6	17.8	.9361	.8852	.8770	.8644	.9785	.9737	.9749	.0057	.9507	.9242	.9157	.9104	.9861	.9818	.9800	.0066

TABLE II

FOURTEEN REPRESENTATIVE MODELS FOR FURTHER EVALUATION ON THE ORSI-4199 [46]. ↑/↓ MEANS A LARGER/SMALLER SCORE IS BETTER. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE, RESPECTIVELY

Methods	ORSI-4199 [46]							
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
VST [37]	.8790	.8717	.8524	.7947	.9481	.9348	.8997	.0281
EGNet [2]	.8464	.8267	.8041	.7650	.9161	.8947	.8620	.0440
PoolNet [30]	.8271	.8010	.7779	.7382	.8964	.8676	.8531	.0541
GateNet [77]	.8680	.8626	.8414	.7946	.9369	.9199	.8816	.0357
SUCA [35]	.8794	.8692	.8590	.8415	.9438	.9356	.9186	.0304
PA-KRN [11]	.8491	.8415	.8324	.8200	.9280	.9168	.9063	.0382
DAFNet [19]	.8552	.8458	.8261	.7819	.9220	.9007	.8905	.0396
EMFINet [21]	.8675	.8584	.8479	.8186	.9340	.9257	.9136	.0330
ERPNet [49]	.8670	.8553	.8374	.8024	.9290	.9149	.9024	.0357
CorrNett [16]	.8623	.8560	.8513	.8534	.9330	.9206	.9142	.0366
ACCoNet [48]	.8761	.8657	.8591	.8562	.9398	.9297	.9163	.0314
MJRBM [46]	.8593	.8493	.8309	.7995	.9311	.9102	.8891	.0374
ASTT [50]	.8828	.8788	.8673	.8527	.9512	.9435	.9216	.0273
Ours	.8811	.8811	.8763	.8758	.9492	.9428	.9261	.0284

predominant, and the presence of multiscale salient objects is a recurrent phenomenon. Consequently, our initial scene was designed to address this multiscale challenge. From the visuals, it is evident that most models can accurately identify objects that should be salient. However, traditional NSI-SOD and ORSI-SOD models such as VST, SMFF, and AGNet falter in precisely pinpointing the object's specific positions, resulting in instances of missed detection. In contrast, RAGRNet mitigates the challenges of multiscale attributes by adeptly highlighting minute objects. Even in scenarios as intricate as the tail fins of aircraft, RAGRNet expertly delineates object contours, a capability absent in other models, as observed in the third instance involving AESINet.

In the second scenario, we present the detection effects in low-light conditions, which can be categorized primarily into two types: objects sharing highly similar colors with their surroundings and objects affected by external factors like heavy fog, common occurrences in practical scenarios. Overall, most methods capture the saliency of objects and approximately identify their locations. Nonetheless, certain models exhibit limitations. For instance, HFANet and F3Net make errors in detecting the first instance of a volleyball court, overlooking one of the objects. Similarly, CNN-based models like AESINet, BMGT, and BAFSNet encounter similar challenges in the second instance. Despite the inherent complexity of these scenarios, RAGRNet effectively tracks object positions, yielding impressive outcomes. While ASTT also accomplishes this task, our model holds a boundary in contour delineation, resulting in a closer alignment with real-world images. Moreover, in the third instance, dense fog obscures a section of an aircraft, leading some models to incorrectly infer a fragmented aircraft. However, our model accurately discerns the complete object, highlighting significant features like side-wing propellers.

The intricate nature of artificial environments exacerbates the challenges in object saliency detection, as depicted in the third scenario. In this context, influenced by the ambient surroundings, certain models (e.g., RADF, PA-KRN, and AGNet) erroneously identify nearby buildings as salient objects. In contrast, RAGRNet overcomes these obstacles by delving into the inherent relationships within objects, achieving precise saliency representation concerning boundary and region relationships that closely mirror real images.

The challenge posed by object topology is showcased in the fourth scenario. Conventional CNN-based methods grapple with capturing global features due to elongated and complex structures, causing the omission of regional information and

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resulting in incomplete object representations. For instance, PoolNet, F3Net, and AESINet exhibit errors in the continuity of boundary information due to internal color variations, leading to overall performance limitations. Existing methods often excel with regular objects but lack the capacity to learn from irregular objects, such as the curved object in the second instance, where models like BMGT, BAFSNet, and HFANet struggle with information enhancement. Transformer-based approaches like VST face challenges in comprehensive global information capture due to inadequate topological learning. Conversely, our proposed RAGRNet successfully addresses these concerns, showcasing satisfactory performance in these aspects.

The final scenario encapsulates the challenges from the previous scenarios, and irregular topology objects, influenced by their surroundings, prove to be a significant puzzle for models to differentiate from the background. For example, in the first instance, most models struggle to distinguish the concave area in the middle of the aircraft. In the second example, elongated and unevenly wide river branches complicate segmentation, causing certain models to inaccurately merge them into a single entity, as seen in PA-KRN, PoolNet, and AESINet. Likewise, in the third instance, RAGRNet adeptly distinguishes the thin, irregular river, overcoming background complexity. In conclusion, our model excels in perceiving global context and object relevance.

3) Computational Efficiency Analysis: We have conducted a comprehensive assessment of computational complexity based on three pivotal metrics: Params, FLOPs, and FPS processing rate in Table I. Analyzing the data, a distinct pattern emerges, showcasing our model's significant superiority in terms of network parameter count and FLOPs compared to numerous benchmark methods. Specifically, our model requires a mere 35.6M Params and entails only 17.82G FLOPs. In the context of FPS, the majority of methods maintain performance within 30–40 FPS, and our approach achieves a real-time inference rate of 35.6 FPS. The comprehensive evaluation results underscore the notable computational efficiency advantage of our model over other high-performing methods.

4) Attribute-Based Analysis On ORSI-4199: The ORSI-4199 dataset comprises nine challenging attribute scenarios: BSO, CS, CSO, ISO, LCO, MSO, NSO, OC, and SSO. We conducted further performance evaluation on these attributes, using the F_β as an example in Table III. It secures the top rank in seven out of the nine attributes, only falling to second place in CSO and ISO. Moreover, it maintains this superiority in terms of average performance as well. In comparison to the second-ranked ASTT, with an average improvement of 0.01, our model demonstrates a significant advancement. These findings unequivocally demonstrate RAGRNet's exceptional performance across a diverse range of challenging scenarios, highlighting its remarkable robustness. However, on the flip side, our model exhibits suboptimal performance in discerning incomplete objects under the ISO attribute, indicating substantial room for improvement in this aspect. Addressing this concern requires dedicated attention and further optimization to enhance the model's performance when dealing with incomplete objects.

TABLE III

ATTRIBUTE-DRIVEN PERFORMANCE ANALYSIS ON THE ORSI-4199 [46]. THE CORRESPONDING MAX F_β SCORES FOR SPECIFIC ATTRIBUTES ARE SHOWCASED. THE AVERAGE OUTCOMES FOR NINE ATTRIBUTES ARE CONSOLIDATED IN THE AVG ROW, WITH THE TOP THREE SCORES FOR EACH ATTRIBUTE HIGHLIGHTED IN RED, GREEN, AND BLUE, RESPECTIVELY

Attr	PoolNet [30]	EGNet [2]	MJRBM [46]	CorrNet [16]	ACCoNet [48]	ASTT [50]	Ours
BSO	.8833	.9139	.9166	.8891	.9091	.9180	.9249
CS	.8603	.8734	.8961	.8835	.8974	.9090	.9105
CSO	.8539	.8705	.8899	.8728	.8931	.8900	.8918
ISO	.8643	.9017	.8937	.8742	.8997	.9230	.9167
LSO	.7605	.7808	.7916	.7852	.8048	.7850	.8202
MSO	.8324	.8372	.8585	.8528	.8611	.8570	.8727
NSO	.8253	.8479	.8748	.8724	.8814	.8980	.8988
OC	.8060	.7897	.8422	.8409	.8337	.8370	.8483
SSO	.7755	.7681	.8091	.8051	.8065	.8000	.8236
Avg	.8291	.8426	.8636	.8529	.8652	.8686	.8786

C. Ablation Studies and Related Discussions

In this section, an extensive array of experiments was conducted on the EORSSD and ORSSD benchmarks to unveil the quantitative and qualitative effects of each component within the proposed framework.

1) Effectiveness of the Individual Modules: To validate the contributions of MGBF, RAGR, and HSCI within RAGRNet, a series of ablation experiments were conducted, and the results are summarized in Table IV. Initially, we established a baseline model (referred to as BL), comprising a Res2Net-based encoder and three convolutional layers for decoding. Subsequently, each component was introduced incrementally, altering one aspect at a time for systematic exploration. As deduced from the results in Table IV, the baseline exhibited the poorest overall performance, aligning with our expectations. Upon the integration of MGBF, RAGR, and HSCI, there was a noticeable enhancement in model performance. Specifically, the introduction of these components led to respective increases in S_α of 0.0084, 0.0095, and 0.0093, as well as corresponding $\mathcal{M}$ improvements of 0.0008, 0.0012, and 0.0014 in EORSSD. The combination of MGBF, RAGR, and HSCI further boosted performance, evident in metrics like $F_\beta^{\max}$: 0.8803 (No.7) versus 0.8765 (No.3) versus 0.8776 (No.4). In comparison with BL, the comprehensive RAGRNet with all components showcased a performance boost of 0.0118 in $E_\xi^{\max}$. Similar trends were evident on ORSSD. In terms of $\mathcal{M}$, RAGRNet exhibited a performance increase of 0.0044 over BL. Comparatively, concerning $E_\xi^{\max}$, the presence of MGBF, RAGR, and HSCI components led to performance improvements of 0.0081, 0.0064, and 0.0043, respectively. Through the scene examples depicted in Fig. 10, we qualitatively illustrated the distinctions. It is evident that model effectiveness was inferior when any component was omitted, as compared to the collaborative performance when all components were present. Notably, in the absence of MGBF, the saliency maps lacked intricate contour information. The lack of inferential relations without RAGR caused the omission of salient information, rendering accurate localization elusive. Similarly, the absence of HSCI hindered the successful attainment of desired results due to the lack of region boundary information.

2) Effectiveness of Different Role in the RAGR: To probe into the impact of the graph reasoning module on the

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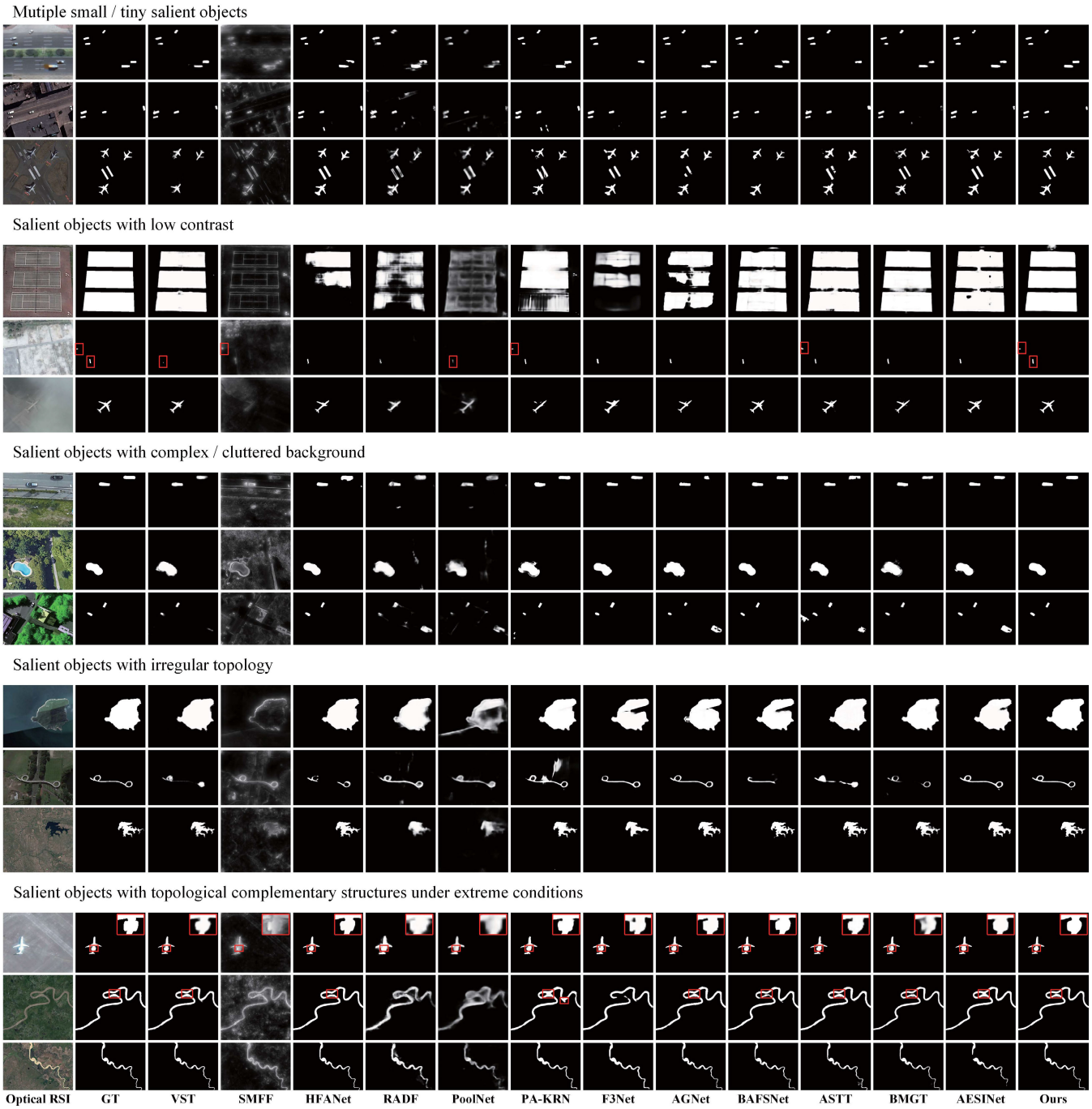


Fig. 9. Visual comparison of different saliency models, including one traditional NSI-SOD model (VST [37]), four CNN-based NSI-SOD models (RADF [76], PoolNet [30], PA-KRN [11], and F3Net [10]), two traditional RSI-SOD models (SMFF [42] and HFANet [79]), three CNN-based RSI-SOD models (AGNet [81], BAFSNet [68], and AESINet [69]), and two transformer-based RSI-SOD methods (ASTT [50] and BMGT [70]). Regions that are challenging to distinguish have been highlighted using red border annotations. In particular, certain regions are magnified for a detailed examination of the detection performance.

performance enhancement of RAGRNet, we conducted a thorough analysis of its iteration count, with the specific outcomes meticulously compiled in Table V. It becomes evident that in the absence of the integrated graph reasoning module, the performance notably lagged behind, as evidenced by both EORSSD and ORSSD. However, upon the integration of the graph reasoning module, there was a gradual rise in performance, which was particularly pronounced when $N = 2$. For instance, with $N = 2$, the S_α on EORSSD saw an increase of 0.0050, while the $E_\xi^{\max}$ observed an augmentation of 0.0053; similarly, on the ORSSD, the $F_\beta^{\max}$ displayed an improvement of 0.0098. It is noteworthy that as N exceeded 2, such as $N = 3$, a performance decline became apparent, particularly on ORSSD, where the $\mathcal{M}$ even ranked last among all comparative results, and the $E_\xi^{\max}$ decreased by 0.0079. Furthermore, we delved into the contribution of CNN local features in

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TABLE IV

ABLATION STUDY ON EVALUATING THE INDIVIDUAL CONTRIBUTION OF EACH CONTENT IN RAGRNET, INCLUDING MGBF, RAGR, AND HSCI. THE BEST RESULT IS INDICATED BY BOLD, THE SECOND RESULT IS INDICATED BY SLANTING, AND THE THIRD RESULT IS INDICATED BY USING UNDERSCORES

No.	Baseline	MGBF	RAGR	HSCI	EORSSD [19]				ORSSD [12]			
					$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$\mathcal{M}\downarrow$
1	✓				.9218	.8646	.9667	.0074	.9277	.8981	.9724	.0110
2	✓	✓			.9302	.8768	.9750	.0066	.9299	.9068	.9758	.0099
3	✓		✓		.9313	.8765	.9748	.0062	.9336	.8992	.9763	.0093
4	✓			✓	.9311	.8776	.9757	*.0060*	.9402	.9090	.9782	.0084
5	✓	✓	✓		.9327	.8769	.9750	**.0057**	.9380	.9079	.9780	.0083
6	✓	✓		✓	.9186	.8650	.9673	.0072	.9429	.9112	.9797	.0081
7	✓		✓	✓	**.9385**	*.8803*	*.9760*	.0061	*.9442*	*.9155*	*.9818*	*.0078*
8	✓	✓	✓	✓	*.9361*	**.8852**	**.9785**	**.0057**	**.9507**	**.9242**	**.9861**	**.0066**

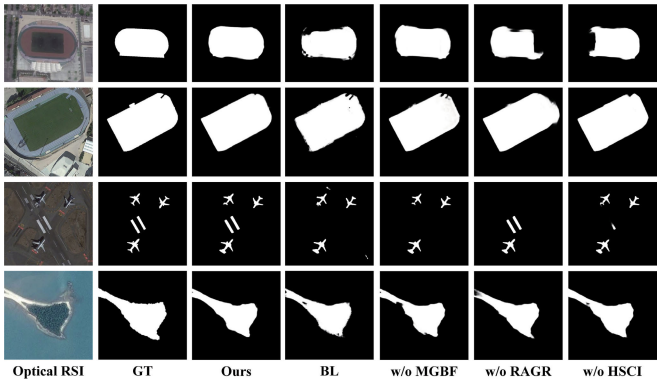


Fig. 10. Visual comparison results of the individual contribution of baseline, MGBF, RAGR, and HSCI.

TABLE V

ABLATIVE INVESTIGATION WAS CONDUCTED ON THE ITERATION COUNT OF THE RAGR FRAMEWORK OVER THE EORSSD [19] AND ORSSD [12]. THE BEST RESULT IN EACH COLUMN IS BOLD

	EORSSD [19]				ORSSD [12]			
	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$\mathcal{M}\downarrow$
N=0	.9311	.8801	.9731	.0060	.9459	.9144	.9798	.0077
N=1	.9335	.8798	.9764	.0059	.9445	.9153	.9812	.0077
N=2	**.9361**	**.8852**	**.9785**	**.0057**	**.9507**	**.9242**	**.9861**	**.0066**
N=3	.9354	.8851	.9772	.0058	.9480	.9198	.9782	.0081

enhancing RAGRNet, the results of which are encapsulated in Table VI. Without the bolstering presence of CNN Local-Guided features, the performance experienced varying degrees of reduction on S_α, as indicated by the figures 0.9361 (with CNN) versus 0.9339 (w/o CNN) for EORSSD and 0.9507 (with CNN) versus 0.9465 (w/o CNN) for ORSSD. To offer a more intuitive depiction of the outcomes of this ablation study, we present qualitative results in Fig. 11. Particularly for the intricacies of the first and second examples, especially the regions marked in red, the introduction of $N = 2$ brought about Precision in outlining the boundaries, presenting a closer semblance to the actual image compared to alternative delineation methods. Furthermore, augmented by the assistance of CNN local features, these regions manifested a heightened sense of saliency, effectively capturing the distinguishing traits of the respective areas.

3) Contributability of Different Actors in HSCI: To assess the effectiveness of each conjecture introduced in HSCI, we conducted an extensive array of ablation experiments.

TABLE VI

ABLATION STUDIES FOR EXPLORING THE COMPLEMENTARY ROLE OF CNN LOCAL FEATURES IN ASSISTING RAGR. THE BEST RESULT IN EACH COLUMN IS BOLD

Varient	EORSSD [19]				ORSSD [12]			
	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$\mathcal{M}\downarrow$	$S_\alpha\uparrow$	$F_\beta^{max}\uparrow$	$E_\xi^{max}\uparrow$	$\mathcal{M}\downarrow$
w/o CNN	.9339	**.8859**	**.9789**	.0062	.9465	.9210	.9836	.0071
with CNN	**.9361**	.8852	.9785	**.0057**	**.9507**	**.9242**	**.9861**	**.0066**

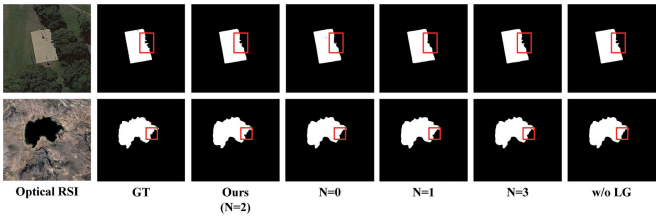


Fig. 11. Visual comparison results of the iteration count of the RAGR framework and the complementary role of CNN local features in assisting RAGR.

These experiments involved dissecting the HSCI to isolate and examine the impact of individual components. Table VII presents the results of our inquiry into MRA, GIA, CPM, and the interaction between boundary and region (referred to as Intact). From Table VII, it is evident that modifying the internal architecture of HSCI leads to a more or less noticeable decline in model performance. When lacking components for enhancing regional information, specifically MRA and GIA, the metrics S_α on the EORSSD exhibit reductions of 0.0018 and 0.0038, respectively. Similarly, the absence of CPM, which is responsible for refining regional and boundary information, results in a decrease of 0.0163 in $F_\beta^{\max}$ on the ORSSD. Furthermore, under the condition of retaining MRA, GIA, and CPM, the sole omission of interaction between boundary and regional information also leads to performance degradation. This observation indirectly suggests that the complementary iteration of region and boundary information contributes to more effective capture of salient features. In addition, Table VIII provides insights into the consequences of omitting channel splitting (referred to as CSplit) and the integration of Multidimensional Unbalanced convolution groups (referred to as UBConv). Analyzing Table VIII, the omission of CSplit corresponded to performance declines, with $F_\beta^{\max}$ on EORSSD and ORSSD dropping by 0.003 and 0.0088, respectively. Moreover, the exclusive of CSplit in the decoding process increased the model's parameter count by 9.08M, underscoring its dual advantage of enhancing performance while controlling parameter complexity. Furthermore, the introduction of UBConv demonstrated discernible benefits. For instance, there were improvements of 0.001 and 0.006 in $E_\xi^{\max}$ scores for EORSSD and ORSSD, respectively. The outcomes of these experiments are visually depicted in Fig. 12. Remarkably, the salient objects within the depicted scenes were intricately intertwined with their surroundings, exacerbating the overall challenges of the detection process. The absence of multilevel boundary and region information from MRA, GIA, and CPM led to incomplete object delineation or hindered precise localization of salient objects, resulting in

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TABLE VII

ABLATION STUDY OF MRA, GIA, CPM, AND INTACT IN HSCI. THE BEST RESULT IN EACH COLUMN IS BOLD

Varient	EORSSD [19]				ORSSD [12]			
	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
w/o MRA	.9343	.8821	.9774	**.0057**	.9402	.9110	.9778	.0087
w/o GIA	.9323	.8779	.9759	.0061	.9409	.9131	.9795	.0085
w/o CPM	.9250	.8641	.9672	.0065	.9415	.9079	.9773	.0078
w/o Intact	.9321	.8814	.9759	**.0057**	.9414	.9133	.9786	.0085
Ours	**.9361**	**.8852**	**.9785**	**.0057**	**.9507**	**.9242**	**.9861**	**.0066**

TABLE VIII

ABLATION STUDY OF CSPLIT AND UBCONV IN HSCI. THE BEST RESULT IN EACH COLUMN IS BOLD

Varient	Params	EORSSD [19]				ORSSD [12]			
		$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$E_\xi^{max} \uparrow$	$\mathcal{M} \downarrow$
w/o CSplit	44.68M	**.9337**	.8822	.9771	**.0057**	.9397	.9154	.9796	.0085
w/o UBConv	**35.13M**	.9331	.8822	.9775	.0061	.9435	.9155	.9801	.0078
Ours	35.60M	.9361	**.8852**	**.9785**	**.0057**	**.9507**	**.9242**	**.9861**	**.0066**

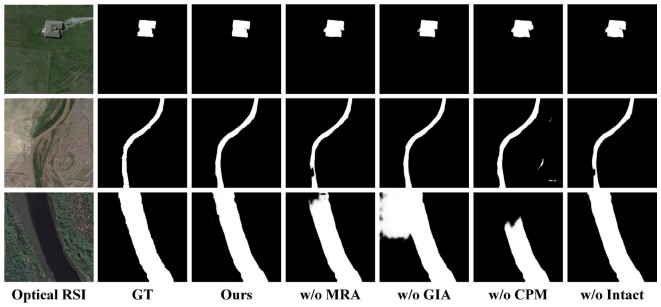


Fig. 12. Visualization of the contributions of MRA, GIA, CPM, and Intact in HSCI.

TABLE IX

ABLATION STUDY ON THE EFFECTIVENESS OF L_{BCE}, L_{IoU}, L_{Floss}, AND L_{BCE}^{B}. THE BEST RESULT IN EACH COLUMN IS BOLD

No.	L_{BCE}	L_{IoU}	L_{Floss}	L_{BCE}^{B}	EORSSD [19]				ORSSD [12]			
					$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
1	✓				.9200	.8149	.9297	.0074	.9455	.8922	.9651	.0085
2	✓	✓			.9303	.8632	.9688	.0064	.9477	.8953	.9675	.0075
3	✓	✓	✓		.9266	.8699	.9658	.0061	**.9516**	.9064	.9692	.0070
4	✓	✓	✓	✓	**.9361**	**.8770**	**.9737**	**.0057**	.9507	**.9157**	**.9818**	**.0066**

TABLE X

ABLATION STUDY ON THE HYPERPARAMETER τ, σ, AND λ. THE BEST RESULT IN EACH COLUMN IS BOLD

No.	τ	σ	λ	EORSSD [19]				ORSSD [12]			
				$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$	$S_\alpha \uparrow$	$F_\beta^{mean} \uparrow$	$E_\xi^{mean} \uparrow$	$\mathcal{M} \downarrow$
1	0.5	0.5	0.5	.9239	.8651	.9628	**.0060**	.9495	**.9115**	**.9723**	.0073
2	1.0	1.0	1.0	**.9266**	**.8699**	**.9658**	.0061	**.9516**	.9064	.9692	**.0070**
3	1.5	1.5	1.5	.9124	.8606	.9610	.0064	.9358	.9005	.9631	.0085
4	0.25	0.25	0.50	.9218	**.8677**	**.9630**	**.0060**	.9425	.9039	**.9723**	**.0073**
5	0.25	0.50	0.25	.9212	.8623	.9609	.0062	**.9446**	**.9041**	.9719	.0075
6	0.50	0.25	0.25	**.9219**	.8516	.9604	.0063	.9438	.9030	.9721	.0080
7	0.2	0.2	0.6	**.9237**	**.8629**	.9608	**.0062**	**.9450**	**.9058**	**.9717**	**.0073**
8	0.2	0.6	0.2	.9229	.8583	**.9626**	**.0062**	.9425	.9053	.9715	.0075
9	0.6	0.2	0.2	.9219	.8621	.9616	.0064	.9439	.9035	.9712	.0076

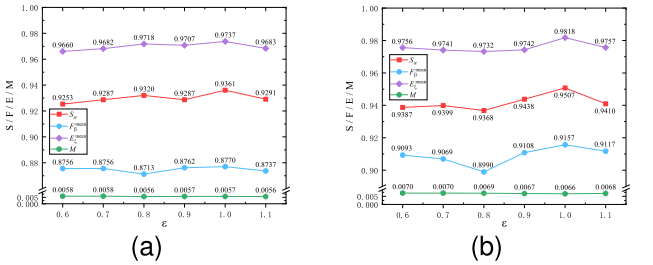


Fig. 13. Performance trends produced with different values of the hyper-parameter ε. The curves of S_α, F_β^{mean}, E_ξ^{mean}, and $\mathcal{M}$ on (a) EORSSD and (b) ORSSD.

deviations during the saliency detection process. In summary, the components devised within the HSCI framework were deemed indispensable, playing a pivotal role in boosting network performance.

4) Loss Ablation: To substantiate the effectiveness of the hybrid loss function, we conducted a comprehensive performance assessment on the trained model, with specific results tabulated in Table IX. The application of the BCE, IoU, and Floss loss functions for the supervision of regional information is, respectively, denoted as L_{BCE}, L_{IoU}, and L_{Floss}. Simultaneously, the BCE loss function utilized for supervising boundary information is designated as L_{BCE}^{B}. As evident from Table IX, the introduction of IoU and Floss losses alongside the BCE loss has led to noticeable enhancements in model performance. It is worth noting, however, that when the BCE loss function is excluded from the boundary information, there is a varying degree of decline in performance metrics compared to the full loss function. This phenomenon indirectly reinforces the efficacy of BCE loss in providing pixel-level supervision and underscores the significance of accurate boundary information extraction within the task.

Continuing, we thoroughly explored the optimal weighting coefficients for L_{BCE}, L_{IoU}, L_{Floss}, and L_{BCE}^{B}. Given the supplementary nature of boundaries as complementary information to regional data, we initially explore the optimal weighting coefficients for τ, σ, and λ as a collective, prioritizing the investigation of their respective contributions, as outlined in Table X. Initially, the coefficients were uniformly set to 0.5, 1, and 1.5 to delineate the influence of τ, σ, and λ across different ranges on the network. Results indicated optimal performance when τ, σ, and λ equaled 1, closely followed by τ, σ, and λ at 0.5, while the least favorable performance was observed with τ, σ, and λ set at 1.5, as presented in rows 1–3 of Table X. Subsequently, we delved into the study of τ, σ, and λ within the 0–1 range to further examine the decisive impact of weighting coefficients, comparing them to row 2 as detailed in rows 4–9 of Table X. Two sets of experiments were conducted, with one maintaining a 1:1:2 ratio for the three factors, and the other adopting a 1:1:3 ratio. In comparison to the optimal τ, σ, and λ value of 1, the overall performance remained slightly suboptimal. Furthermore, building upon τ, σ, and λ set at 1, we further explored the optimal value for ε, as depicted in Fig. 13. Analysis of Fig. 13 indicated a continuous improvement in network performance with an increasing ε, reaching its peak when ε equaled 1. In summary, by amalgamating various loss functions and harmonizing private supervision with saliency supervision, we achieved a satisfactory enhancement in performance and obtained optimal weighting coefficients, with τ, σ, λ, and ε all equaling 1.

5) Network Visualization: To vividly substantiate the efficacy of our proposed modules, we employed feature visualization techniques to showcase the roles of MGBF, RAGR, and HSCI, as illustrated in Fig. 14. Through feature visualization,

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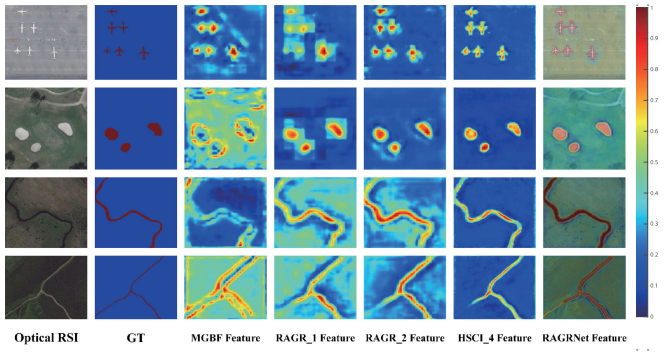


Fig. 14. Feature visualization to show the contribution of each module to feature extraction, including MGBF, RAGR, HSCI, and the whole network.

it was observed that, under the influence of MGBF, the model's attention became concentrated on object boundaries, resulting in a more complete delineation of these boundaries. By overlaying the RAGR modules (with the first layer labeled as RAGR_1 and the second layer as RAGR_2), the network demonstrated strengthened attention focus in localizing irregular object regions. Furthermore, to accommodate the impact of resolution on images, we selected the output features of the final HSCI structure, denoted as HSCI_4, for visualization. In comparison to previous visualization results, the introduction of HSCI significantly enhanced the network's detection performance. Finally, we presented the overall output of RGARNet in a more intuitive manner by mapping it onto the original image, as depicted in the last column, thereby accurately portraying the practical performance of RAGRNet.

D. Analysis of Failure Samples

While our proposed model has demonstrated remarkable performance across various scenarios, there still exist opportunities for further refinement. To emphasize the challenges our model faces, we present a selection of representative failure cases in Fig. 15. In the first case, it becomes evident that recognizing irregular objects accurately, especially amidst scenarios characterized by extreme foreground-background similarity and blurriness, remains a formidable task. This challenge is even more pronounced in the second case. Despite the fact that the graph structure enhances the interplay between regional and boundary information, the presence of preexisting knowledge during the learning process might occasionally lead to erroneous judgments. This is particularly noticeable when distinguishing whether buildings constitute salient regions, as showcased in the third case. In the final example, our model exhibits a degree of difficulty in handling incomplete objects, especially when they are surrounded by interfering elements. Although our model effectively captures the general contour of the authentic image, errors surface in delineating certain intricate boundaries. Importantly, these instances of errors are not unique to our model but are also encountered by other models, including RRNet, which similarly employs graph reasoning.

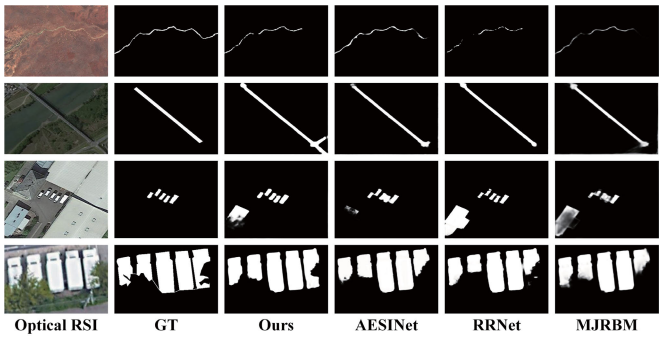


Fig. 15. Failure cases of the proposed RAGRNet, along with AESINet [69], RRNet [20], and MJRBM [46].

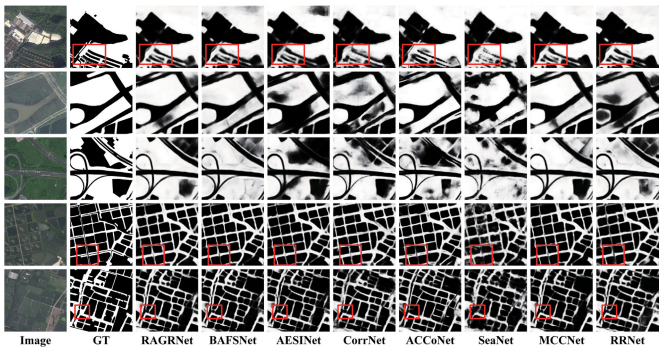


Fig. 16. Visual comparison of different saliency models on UGSet [53], including BAFSNet [68], AESINet [69], CorrNet [16], ACCoNet [48], SeaNet [17], MCCNet [13], RRNet [20], and Ours. Regions that are challenging to distinguish have been highlighted using red border annotations.

E. Generalization in Large-Area Detection

To assess the proposed model's generalizability on a large-area dataset, we conducted retraining on UGSet [53], along with the selection of seven classical networks. The results of these networks are presented in Table XI, revealing RAGRNet's superior overall performance, ranking within the top 3 in five metrics, with a first-place position in three of them. Compared to BAFSNet, RAGRNet demonstrated an improvement of 0.0053 in $\mathcal{M}$, becoming a pioneering model as the only one to surpass 0.77 on E_{ξ}^{mean} among all models. In comparison to RRNet, which incorporates graph reasoning networks for performance enhancement, RAGRNet showed a 0.0164 improvement in F_{β}^{adp}. In addition, we showcase the UGS detection results for randomly selected multiclass scenes in Fig. 16. A careful observation of Fig. 16 reveals certain deficiencies in all network models, showing significant disparities from real saliency maps. However, RAGRNet tends to outperform others. For instance, RAGRNet outlines green areas more comprehensively, especially highlighting the unknown area marked by the red box, compared to BAFSNet and RRNet in the first, fourth, and fifth instances. In the second and third instances, RAGRNet exhibits a more comprehensive overall discrimination of green spaces compared to AESINet, CorrNet, and SeaNet, with minimal sensitivity to the varying colors of green space. Overall, RAGRNet, proposed in this study, has room for improvement in large-area saliency detection, particularly in adapting to the diversity of UGS and the grasp of complex relationships between hierarchical levels.

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TABLE XI

EIGHT REPRESENTATIVE MODELS FOR FURTHER EVALUATION ON THE UGSET [53]. ↑/↓ MEANS A LARGER/SMALLER SCORE IS BETTER. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE, RESPECTIVELY

Methods	Publication	UGSet [53]							
		$S_\alpha \uparrow$	$F_\beta^{max} \uparrow$	$F_\beta^{mean} \uparrow$	$F_\beta^{adp} \uparrow$	$E_\xi^{max} \uparrow$	$E_\xi^{mean} \uparrow$	$E_\xi^{adp} \uparrow$	$\mathcal{M} \downarrow$
RRNet [20]	2022 TGRS	.6596	.6438	.6197	.5642	.8130	.7693	.7561	.1429
MCCNet [13]	2022 TGRS	.6464	.6398	.6171	.6223	.8143	.7580	.7576	.1386
SeaNet [17]	2023 TGRS	.6361	.6008	.5712	.5435	.8039	.7502	.7245	.1645
ACCoNet [48]	2023 TCYB	.6536	.6297	.6049	.5946	.8120	.7641	.7449	.1467
CorrNett [16]	2023 TGRS	.6523	.6284	.6002	.5869	.8173	.7631	.7527	.1522
AESINet [69]	2023 TGRS	.6435	.6269	.6050	.5828	.8192	.7684	.7591	.1428
BAFSNet [68]	2023 TGRS	.6627	.6459	.6192	.5697	.8190	.7665	.7579	.1425
Ours	-	.6558	.6399	.6214	.5806	.8121	.7729	.7522	.1372

V. CONCLUSION

In this investigation, we delved deeply into the challenges brought about by the intricate topological structures and diverse scales of artificial salient objects in the domain of ORSI-SOD. To address these challenges, we put forth an innovative RAGRNet that aims to intricately fuse the inherent relationships between region and boundary information. To begin with, we introduced the MGBF, which adeptly amalgamates boundary information of varying granularities, thereby capturing multilevel features and generating finer texture details. Following that, the RAGR was formulated, utilizing self-learned mechanisms to blend global and local contexts, further refining finer details, and enriching the structural information of the target objects. Ultimately, we devised a decoder named HSCI, which facilitated a step-by-step interactive fusion of information, uncovering latent semantic connections that underlie boundaries and regions. Through this approach, we not only extensively probed into the interrelationships among salient objects but also revealed their inherent dependencies. Comprehensive experimental results bear witness to the outstanding performance of our proposed model and the efficacy of each individual component. In the future, our endeavors will shift toward developing more efficient graphic reasoning models capable of effectively navigating through complex scenes with high foreground-background similarity. Furthermore, we will explore the generalization of large-area detection, aiming to maximize its practical value.

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